SAMPLING BIAS IN HOUSEHOLD STUDIES

15 MAY 2024

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OBJECTIVES

Better understand the impact of age in the transmission dynamics in households

Development of a stochastic model of transmission **in households**[1]

Adaptation of the model in collaboration with M. Layan

Data : PedCovid longitudinal study
PI : Pr. Isabelle Sermet-Gaudelus,
Funding : PHRC + ANR

Statistical inference

- Susceptibility
- Infectiousness
- Impact of age on transmission risk

Recruitment through identification of positive children: bias?

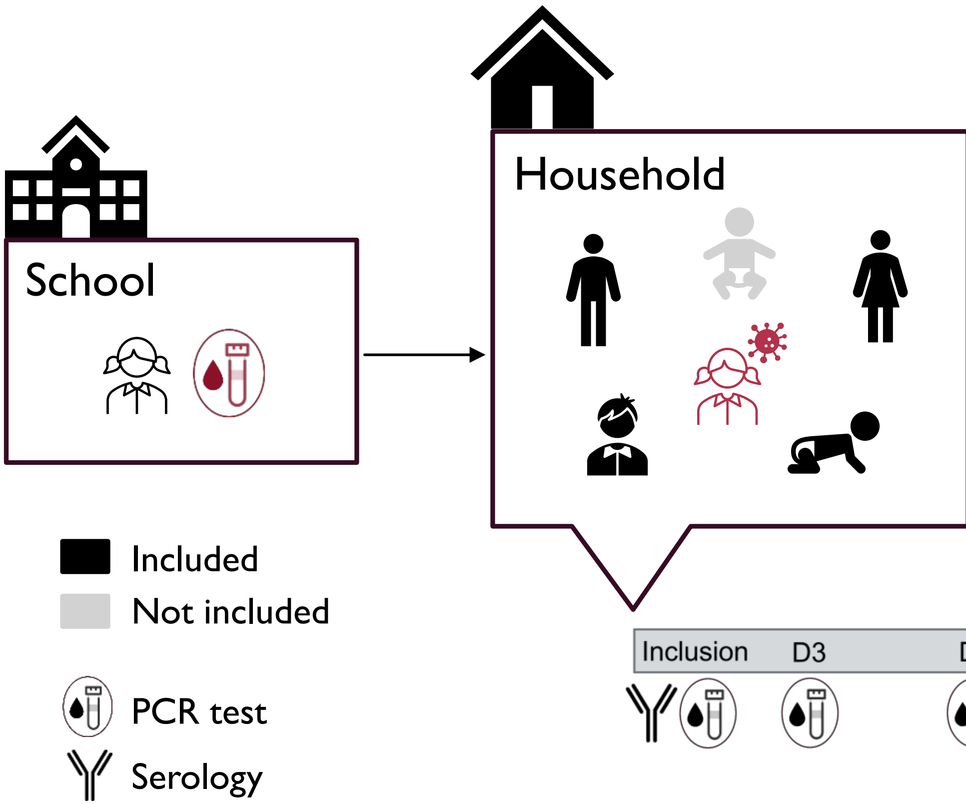
Objective: discuss how to correct/take into account this sampling bias

[1] Layan M, *et al.* Impact of BNT162b2 Vaccination and Isolation on SARS-CoV-2 Transmission in Israeli Households: An Observational Study. *Am J Epidemiol* 2022; **191**: 1224–34.

I – DATA

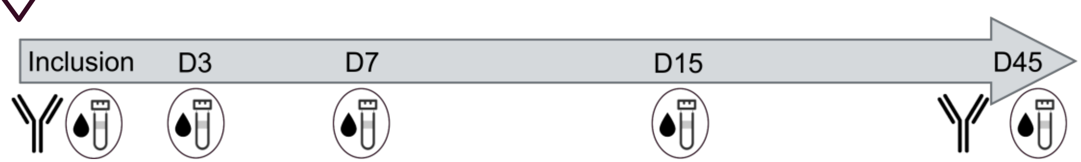
DESIGN
DESCRIPTION

DESIGN



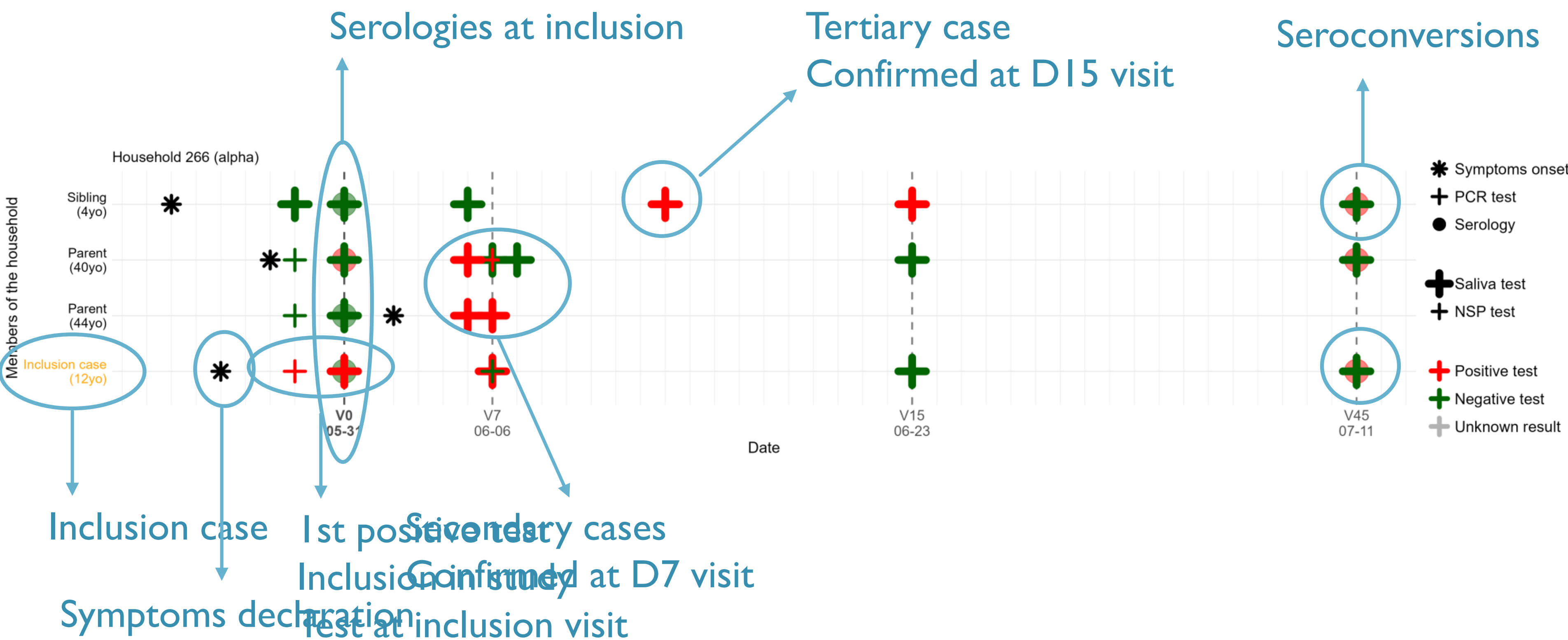
2 recruitment periods :

- 11 March 2021 → 10 June 2021 (ALPHA)
- 14 December 2021 → 17 February 2022 (OMICRON)



- Gender, age
- Vaccination status, previous COVID-19 infections
- Clinical status (symptomatic/asymptomatic)
- Date and type of symptoms when applicable
- Results of serologies and saliva tests

HOUSEHOLD TRAJECTORIES



HOUSEHOLD CHARACTERISTICS

	Alpha period	Omicron period
N	128	54
Household size*	4 (4-5)	4 (4-5)
N with AR=100%	30 (23%)	23 (43%)
Age of inclusion case*	10 (7-13.25)	6 (4.25-8)

*median (IQR)

2 – METHODS

MODEL
STATISTICAL APPROACH

MODEL

- **Estimated parameters of interest:**
  - Relative infectiousness of individuals across ages and symptoms statuses → $\mu_{inf}(s_j, a_j)$
  - Relative suscpetibility of individuals across ages → $\mu_{susc}(a_i)$

- **Age groups:**
  - Infants (<6yo)
  - Children (6-11 yo)
  - Teens/adults (>11yo)

Reference : symptomatic infants (<6 yo)

MODEL

- **Risk of infection of individual i by j at time t :**

$$h_{j\to i}(t) = \frac{\beta}{\left({}^{n_i}/_{4}\right)^{\delta}} \times \mu_{susc}(a_i) \times \mu_{acq}(p_i) \times \mu_{inf}\left(s_j, a_j\right) \times \mu_{transm}\left(p_j\right) \times f\left(t-\tau_j\right)$$

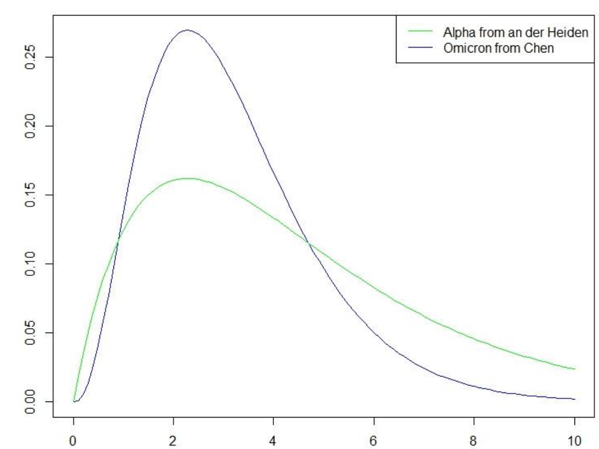
Baseline transmission rate dependent on household size (δ)

Relative susceptibility of infected i :
By age of i (μ_{susc})
By protection of i (μ_{acq})

Relative infectiousness of infector j :
By age and symptomatic status of j (μ_{inf})
By protection of j (μ_{transm})

Likelihood of generation time

Distribution of generation time:



Estimated parameters : $\theta = (\beta, \delta, \mu_s^2, \mu_a, \mu_i^5, \mu_t$

MODEL

- **Risk of infection of individual i at time t :**

$$\lambda_i(t) = \underbrace{\alpha}_{\text{Risk of infection in the community}} + \underbrace{\sum_{j \in H_i, \tau_j < t} h_{j \to i}(t)}_{\text{Risk of infection in the household (individuals from the same houehsold infected before t)}}$$

→ **Likelihood**

→ **Markov Chain Monte Carlo to estimate parameters**

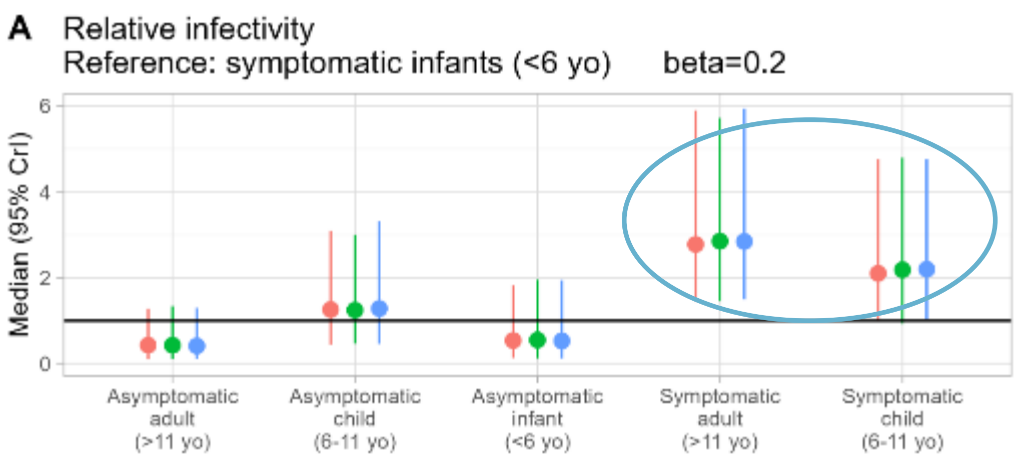
Estimated parameters : $\theta = (\beta, \delta, \mu_s^2, \mu_a, \mu_i^5, \mu_t, \alpha)$

3 – RESULTS

RESULTS

Relative infectiousness

alpha

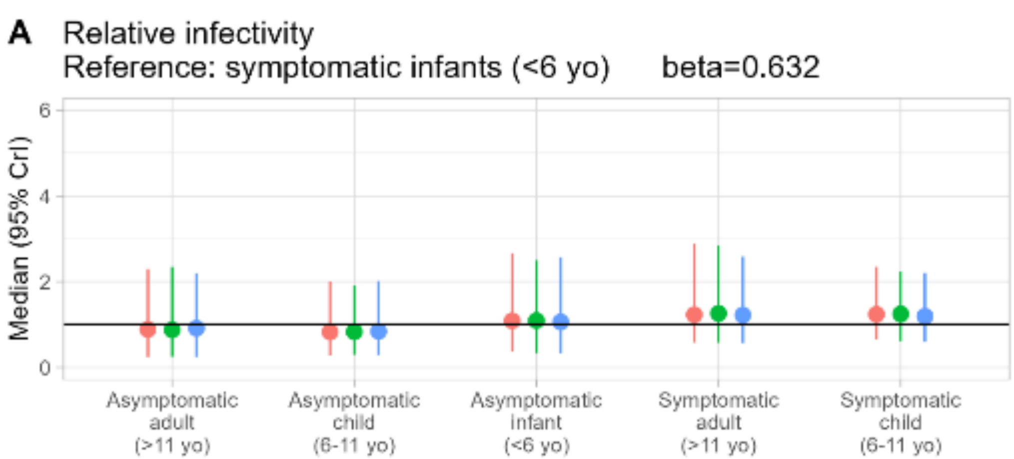


Symptomatic infants (<6 yo)

Alpha:

Among symptomatic individuals:
Infants are less infectious than others

omicron



Symptomatic infants (<6 yo)

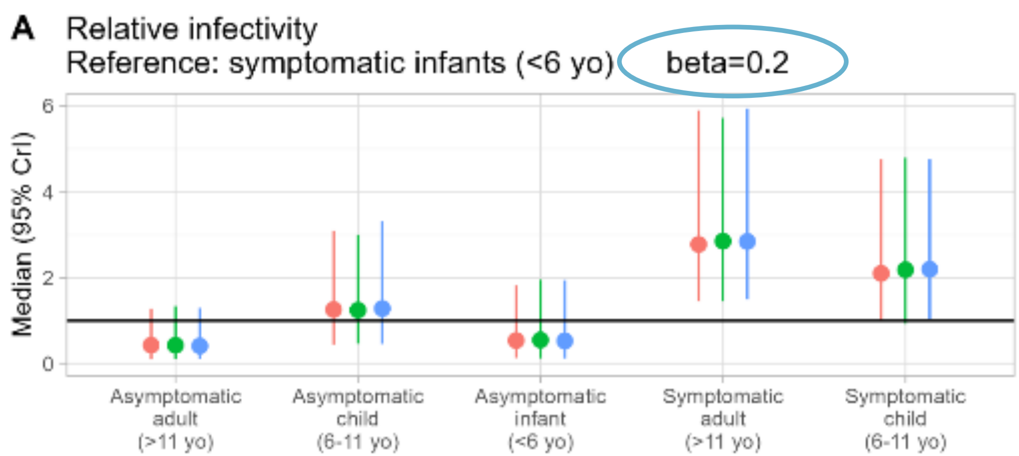
Omicron:

No difference of infectiousness across ages and symptoms (lack of power?)

RESULTS

Relative infectiousness

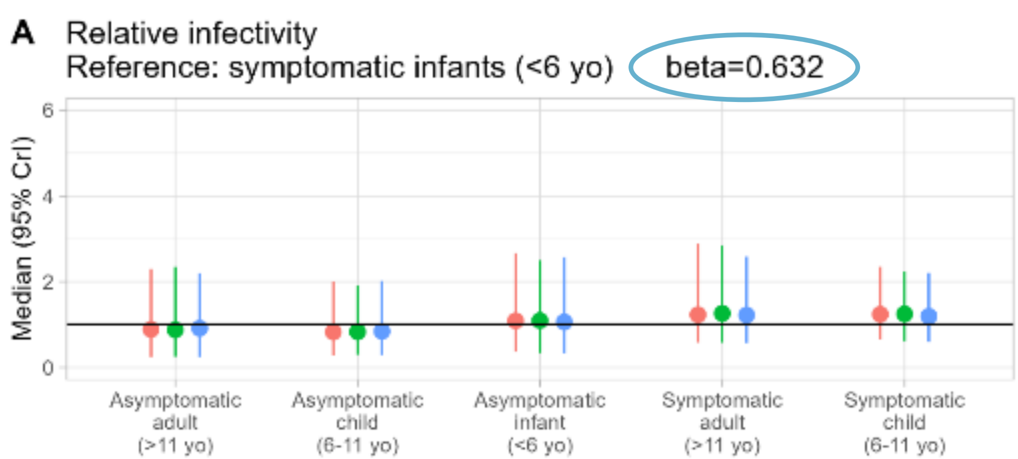
alpha



Symptomatic infants (<6 yo)

Baseline transmission rate higher for Omicron than for Alpha

omicron

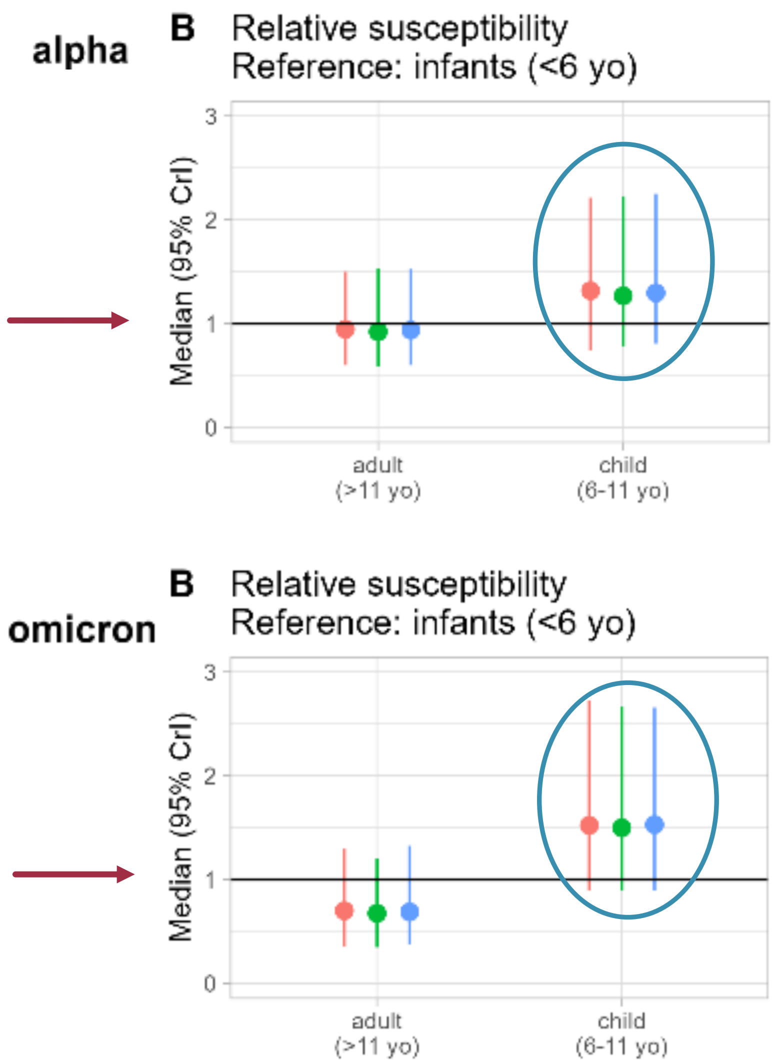


Symptomatic infants (<6 yo)

RESULTS

Relative susceptibility

Infants
(<6 yo)



Infants
(<6 yo)

Tendency:

Children (6-11yo) more susceptible than infants (<6yo)

SUMMARY

Summary :

- Convergence of the method ok
- Some statistically significant results

Limits :

- Limited power (number of households / number of individuals in each age~symptoms category)
- Recruitment of households via children → bias?

→ Simulation study

4 – SIMULATION STUDY

SIMULATION STUDY

Objective : compare scenarios of households recruitment

Fix all parameters

Simulate many household outbreaks

$$\lambda_i(t) = \alpha + \sum_{\substack{j \in H_i, \\ \tau_j < t}} h_{j \to i}(t)$$

?

Include households following a recruitment scenario

Simulated dataset

Stochastic model of household transmission

Statistical inference

Compare parameter estimates to expected values

RECRUITMENT SCENARIOS

Recruitment via **children**
(PedCovid-like)

- If the inclusion case is an adult → exclude
- If the inclusion case is a child → recruit

Random recruitment
(Random)

- No classification/exclusion
- Draw randomly a total number of hh

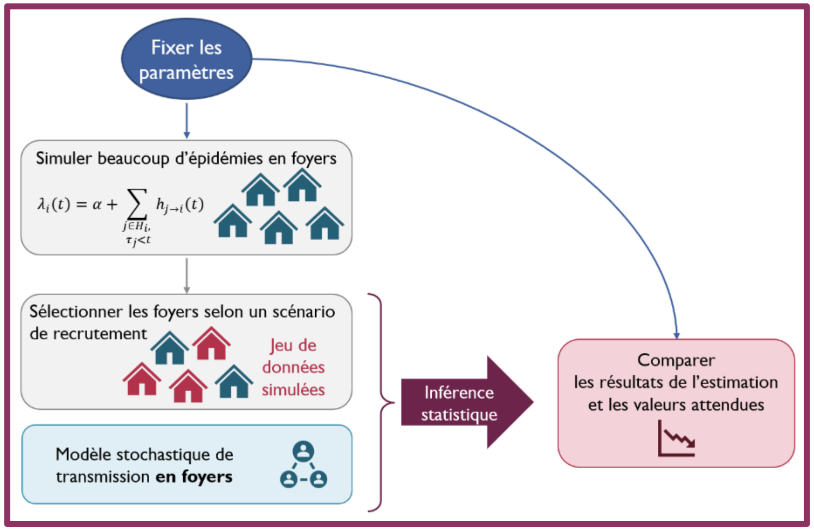
Recruitment via **adults**
(Via adults)

- If the inclusion case is a **child** → exclude
- If the inclusion case is an **adult** → recruit

Over-sampling with recruitment via children
(PedCovid-like + sampling)

- If the inclusion case is an adult → exclude
- **If no adult is infected in the household → exclude**
- If the inclusion case is a child → recruit

SIMULATION STUDY



50 times

For each **recruitment scenario**

And each **parameter set**

Tested parameter sets:

Null hypothesis: no difference in infectiousness/susceptibility across ages or symptoms

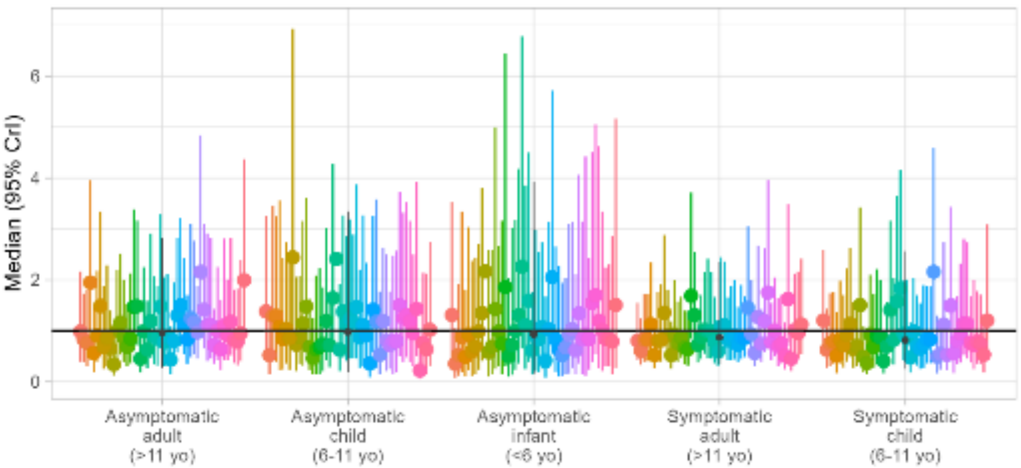
Bias study: vary parameter values around 1 independantly (while the others = 1)

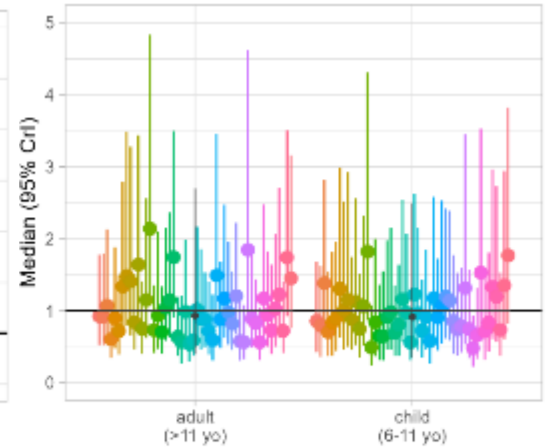
NULL HYPOTHESIS

Relative infectiousness = 1
Relative susceptibility = 1

Random recruitment (alpha)

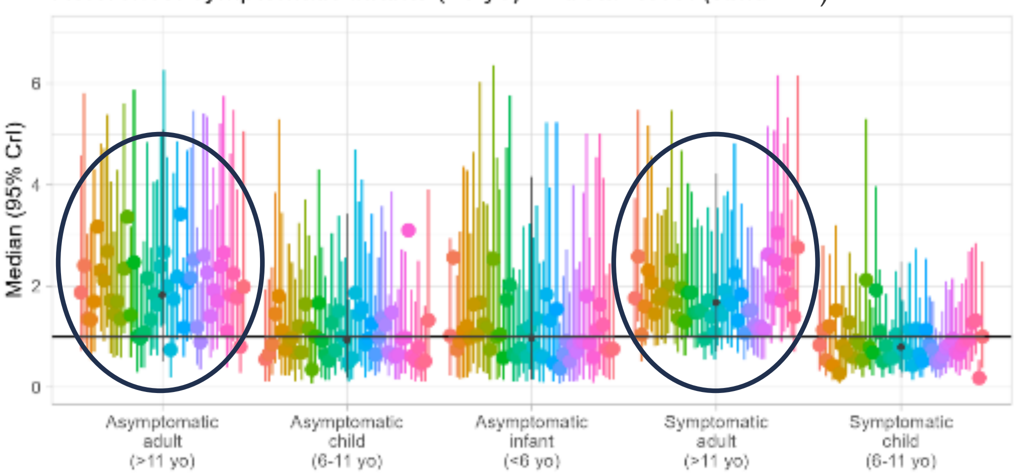
Relative infectivity
Reference: symptomatic infants (<6 yo)   beta=0.204(simu=0.2)



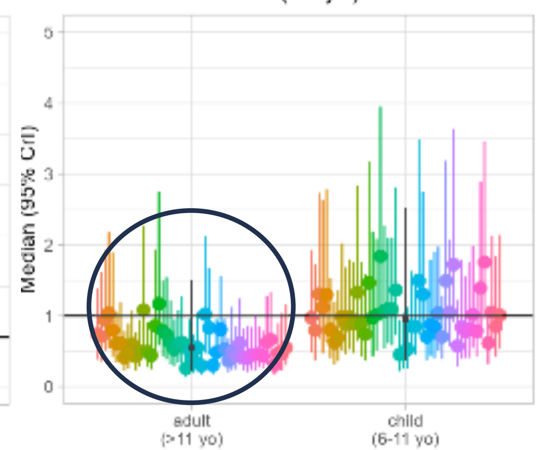


Recrutement PedCovid-like (alpha)

Relative infectivity
Reference: symptomatic infants (<6 yo)   beta=0.187(simu=0.2)



B Relative susceptibility
Reference: infants (<6 yo)



Random recruitment:
Good estimation

PedCovid-like recruitment:
Over-estimation of infectiousness in adults
Under-estimation of susceptibility in adults

NULL HYPOTHESIS

Relative infectiousness = 1
Relative susceptibility = 1

→ Similar results for omicron variant

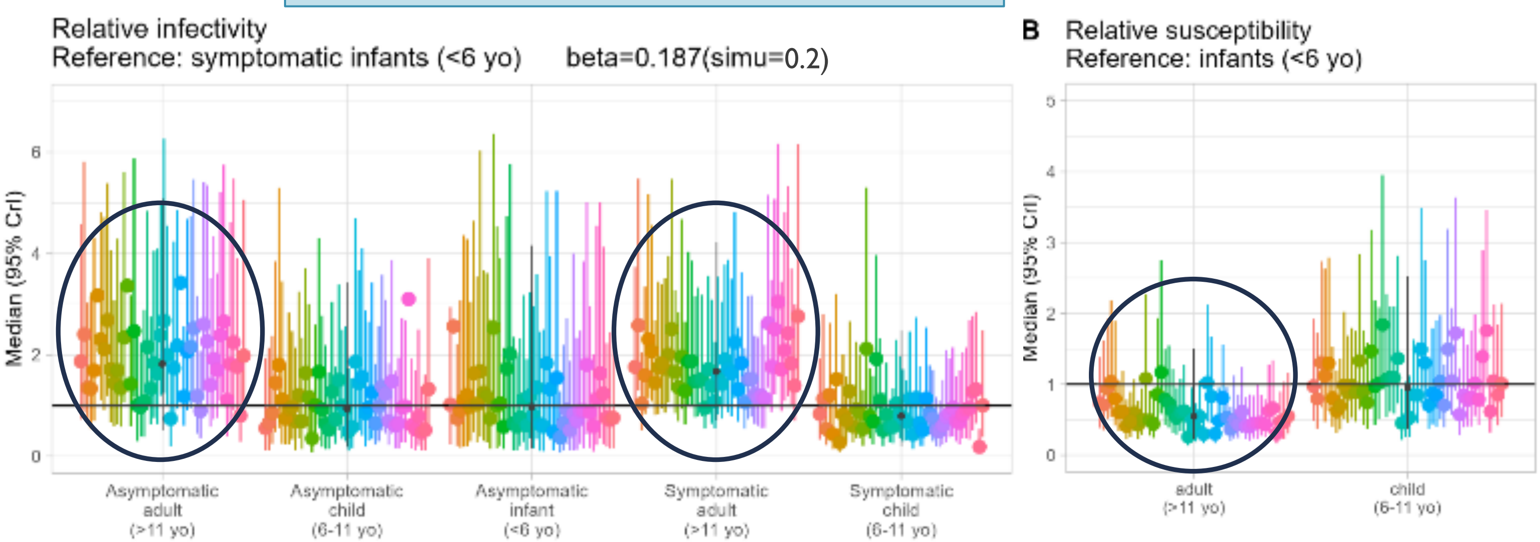
Conclusion :

- Presence of a bias, potentially high and statistically significant
- Test values lower/higher than 1 to evaluate the impact of bias on our results

Random recruitment:
Good estimation

PedCovid-like recruitment:
Over-estimation of infectiousness in adults
Under-estimation of susceptibility in adults

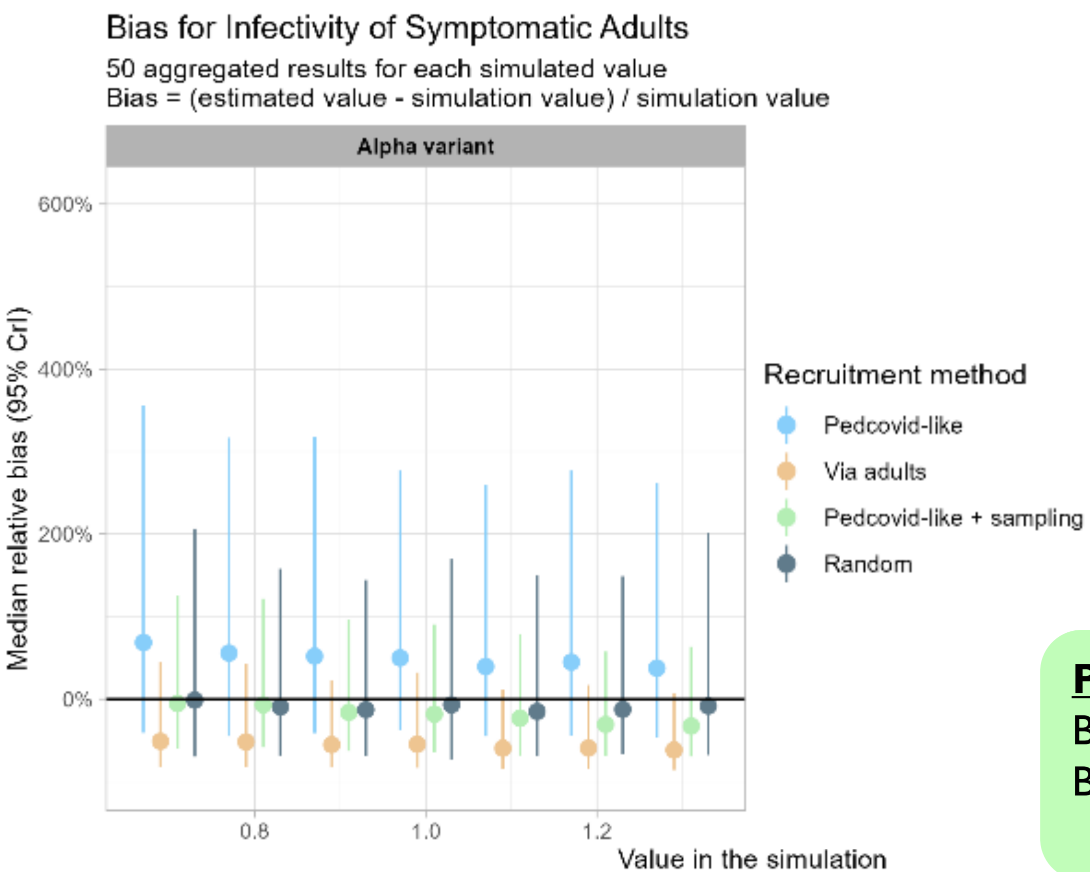
Recrutement PedCovid-like (alpha)



BIAS STUDY

Relative bias = $\frac{estimated\ value - expected\ value}{expected\ value}$ where the expected value is the one used in the simulations

Aggregated results of the 50 simulations for each recruitment scenario:



PedCovid-like recruitment:
Over-estimation of infectiousness in symptomatic adults

Random recruitment:
No apparent bias

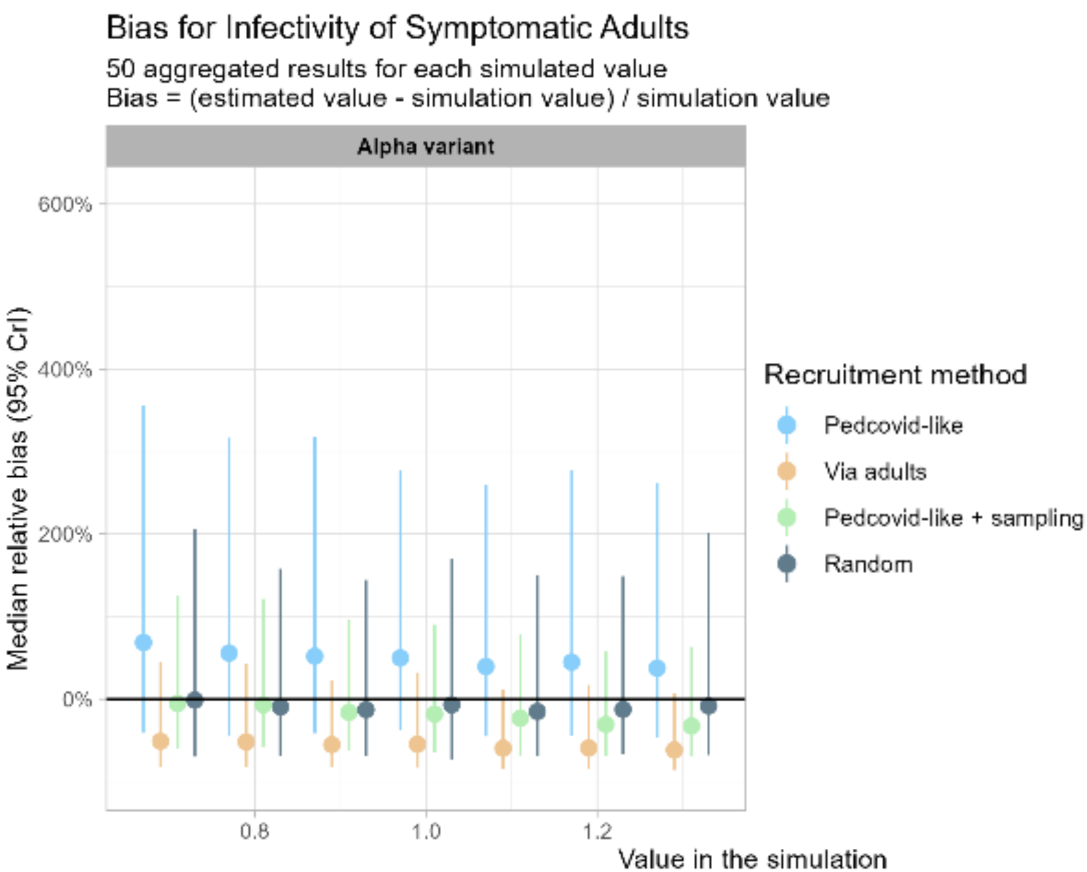
Recruitment via adults:
Under-estimation of infectiousness in symptomatic adultes
→ inverted bias as expected

PedCovid-like recruitment + sampling:
Bias greatly reduced
But decreasing bias
→ probably not a good correction

BIAS STUDY

Relative bias = $\frac{estimated\ value - expected\ value}{expected\ value}$ where the expected value is the one used in the simulations

Aggregated results of the 50 simulations for each recruitment scenario:



→ Similar results for omicron

→ Similar results for the other parameters

CONCLUSION AND PERSPECTIVES

Conclusion:

- Bias whatever the value of the parameter is
- Need to take it into account in the interpretation of results

Perspectives for correction of bias:

- Other sampling method to explore (only hh where inclusion case is index?)
- Directly integrate the recruitment method in the likelihood computation (collaboration with Simon Cauchemez)

- Others?

SUPPLÉMENTS

DATA

CARACTÉRISTIQUES DES INDIVIDUS

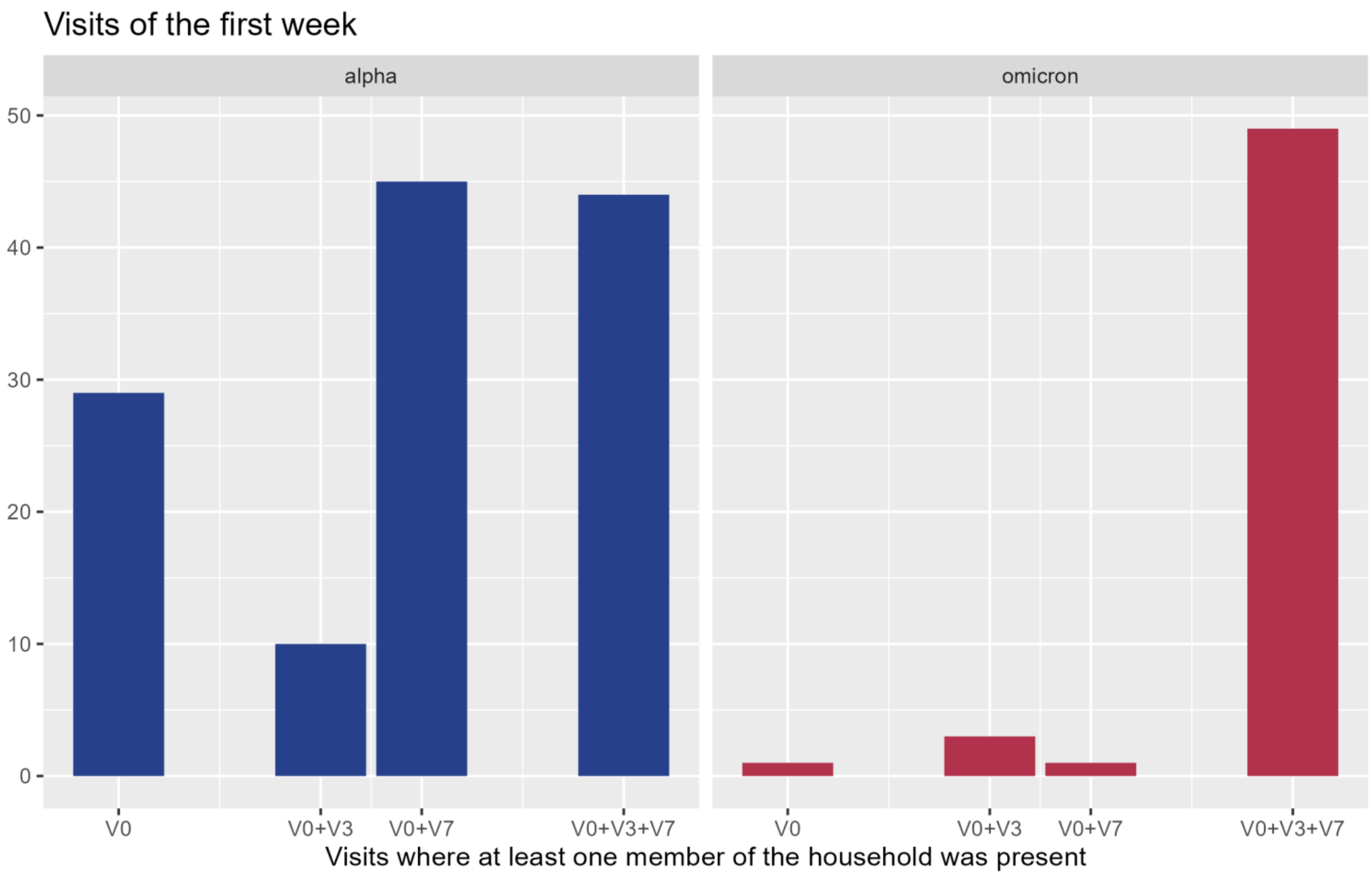
		Alpha period			Omicron period		
N		535			234		
N of positive cases		334 (62%)			179 (76%)		
		Infants (<6yo)	Children (6-11yo)	Teens/adults (>11yo)	Infants (<6yo)	Children (6-11yo)	Teens/adults (>11yo)
N		61	124	350	60	49	125
Sex F		31 (51%)	68 (55%)	187 (53%)	26 (43%)	17 (35%)	61 (49%)
Age		4 (2-5)	9 (7-10)	38 (16-44)	3 (1-5)	8 (7-10)	39 (36-43)
N inclusion cases		21 (16%)	55 (43%)	52 (41%)	24 (44%)	24 (44%)	6 (11%)
Protected	Past infect°	2	1	3	3	3	2
	Vaccinat°	0	1	27	1	2	109(+2)
	None	59	122	320	56	44	14
Infection status	Not infect	23 (38%)	36 (29%)	142 (41%)	11 (18%)	5 (10%)	39 (31%)
	Sympto	28	61	184	39	35	69
	Asympto	10	27	24	10	9	17
Chez les infectés, prévalence asympto:		26%	31%	12%	20%	20%	20%

CARACTÉRISTIQUES DES INDIVIDUS

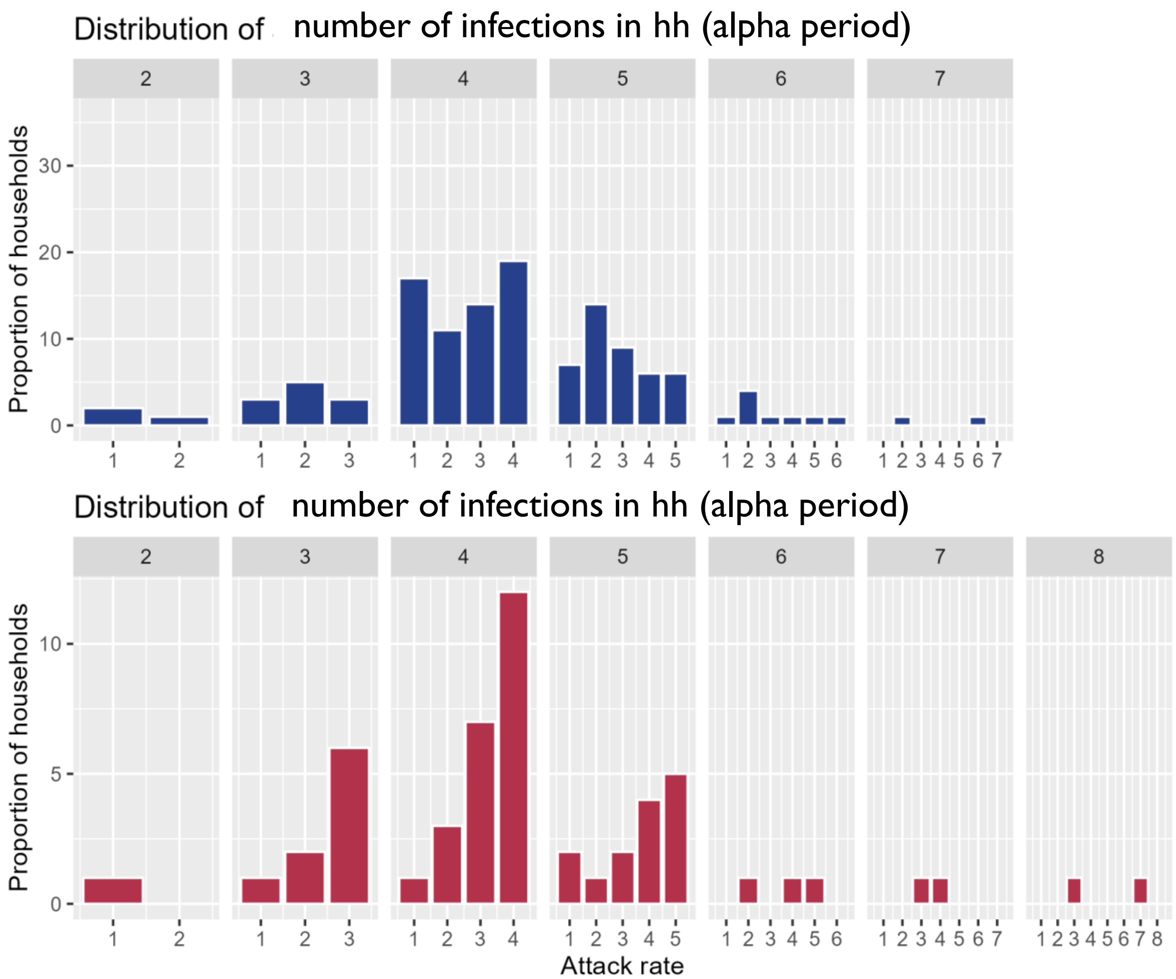
Exclusion des cas index

		Alpha period			Omicron period		
N		407			180		
N of positive cases		206 (51%)			125 (69%)		
		Infants (<6yo)	Children (6-11yo)	Teens/adults (>11yo)	Infants (<6yo)	Children (6-11yo)	Teens/adults (>11yo)
N inclusion cases		21 (16%)	55 (43%)	52 (41%)	24 (44%)	24 (44%)	6 (11%)
N		40	69	298	36	25	119
Sex F		21 (53%)	42 (61%)	159 (53%)	16 (44%)	17 (44%)	61 (50%)
Age		3 (1-5)	9 (7-10)	40 (34-45.75)	2 (1-3.25)	8 (7-10)	39 (36-43)
Protected	Past infect°	1	0	3	3	2	2
	Vaccinat°	0	0	27	0	1	104(+2)
	None	39	69	268	3	22	13
Infection status	Not infect	23 (58%)	36 (52%)	142 (48%)	11 (31%)	5 (20%)	39 (33%)
	Sympto	12	22	138	19	16	63
	Asympto	5	11	18	6	4	17
Chez les infectés, prévalence asympto:		29%	33%	12%	24%	20%	21%

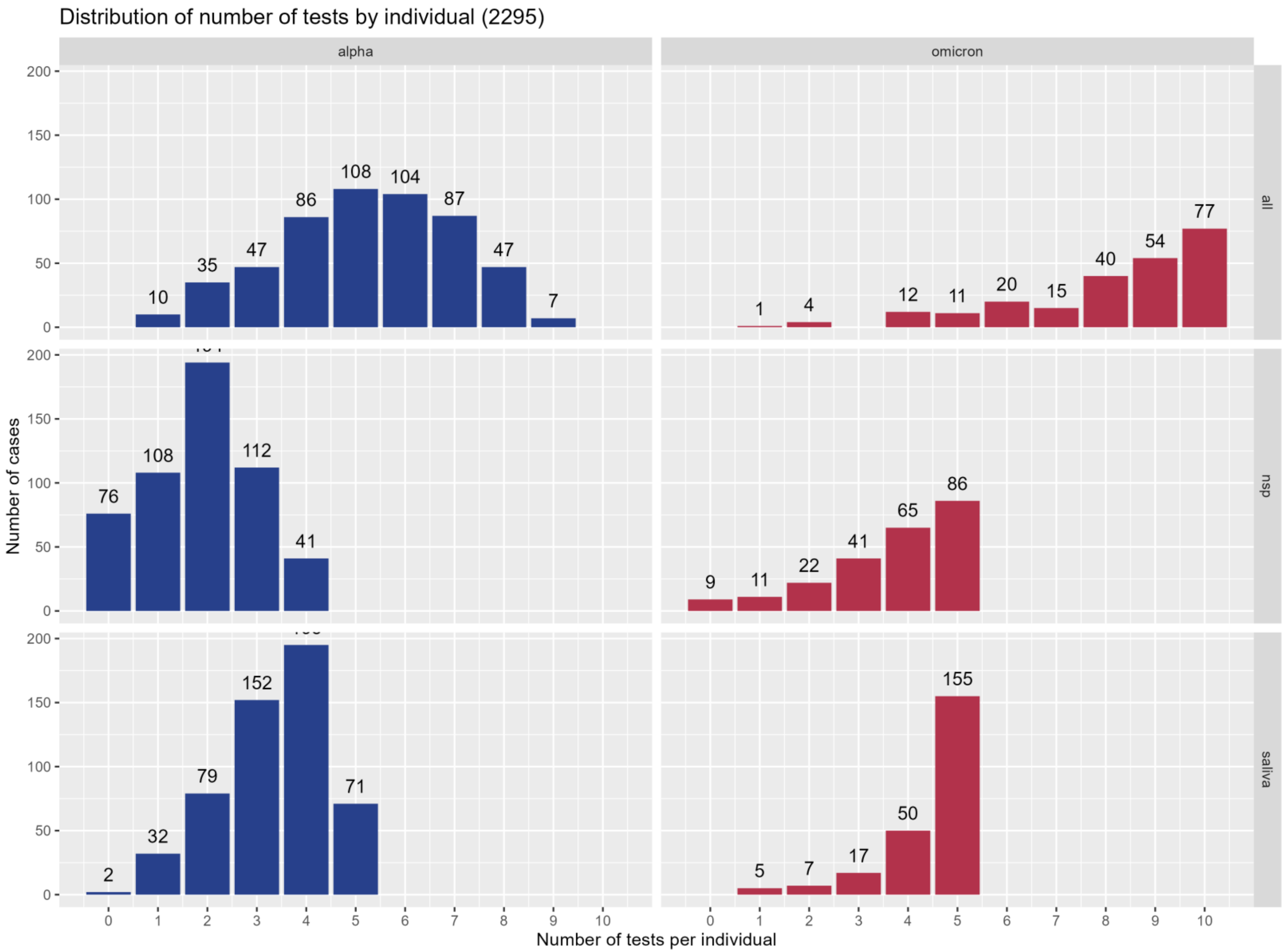
CARACTÉRISTIQUES DES FOYERS



ATTACK RATES



CARACTÉRISTIQUES DES INDIVIDUS



MODEL

HYPOTHESES

- **Data from individual i of household H_i:**
  - Individual id
  - Household id (H_i)
  - Household size (n_i)
  - Symptoms onset date (for symptomatics – **1st positive test** for asymptomatics) (d_i)
  - Infectious status (not infected – symptomatic – asymptomatic) (s_i)
  - Protection status (yes – no) (p_i)
  - Study period (alpha – omicron)

- **Protection**

Un individu est considéré protégé si :

- Infections passées : infection de moins de 6 mois et +30j
- Vaccination : Alpha: 1e dose +10j avant et -90j OU 2e dose +10j avant
  Omicron: 2e dose +10j avant

- **Tranches d'âge**

Bas âge (<6 ans)

Enfance (6-11 ans)

Ados/adultes (12+ ans)

MODEL

- **Data augmentation:** unobserved infection date (τ_i):

Symptoms onset date/1st test date – incubation period (distribution g)

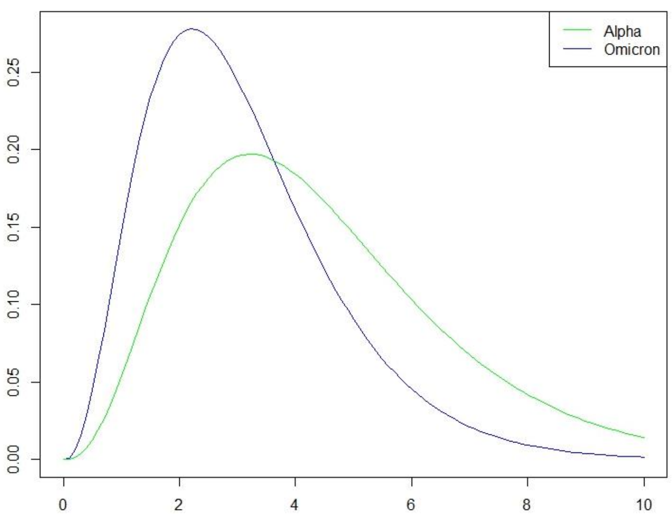
Distribution of incubation period:

Symptomatic : distributions from S. Galmiche et al.

ALPHA : gamma(mean=4.42, sd=2.3)
OMICRON : gamma(mean=3.09, sd=1.64)

Asymptomatic :

ALPHA : Unif(2,7)
OMICRON : Unif(1,5)



MODEL

- **Risk of infection of individual i by j at time t :**

$$h_{j\to i}(t) = \frac{\beta}{\left({n_i}/{4}\right)^{\delta}} \times \mu_{susc}(a_i) \times \mu_{acq}(p_i) \times \mu_{inf}(s_j, a_j) \times \mu_{transm}(p_j) \times f(t-\tau_j)$$

Baseline transmission rate dependent on household size (δ)

Relative susceptibility of infected i :
By age of i (μ_{susc})
By protection of i (μ_{acq})

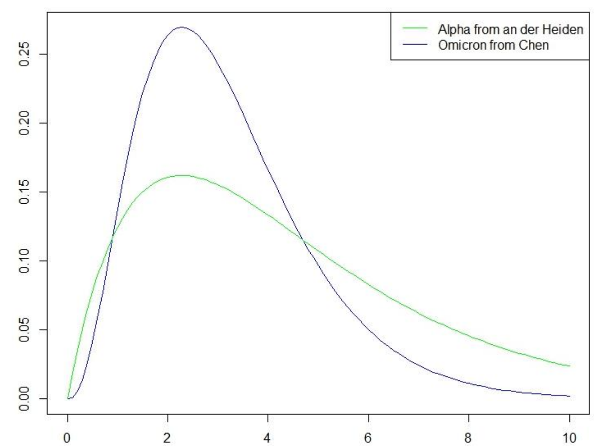
Relative infectiousness of infector j :
By age and symptomatic status of j (μ_{inf})
By protection of j (μ_{transm})

Likelihood of generation time

Distribution of generation time:

ALPHA : distribution from an der Heiden et al.
→ gamma(shape=2, rate=0.44)

OMICRON : distribution from Chen et al.
→ gamma(shape=3,531, rate=1.098)



Estimated parameters : $\theta = (\beta, \delta, \mu_s^2, \mu_a, \mu_i^5, \mu_t$

MODEL

- Estimated parameters: $\theta = (\beta, \delta, \mu_s^2, \mu_a, \mu_i^5, \mu_t, \alpha)$
- **Likelihood of the data:**

$$\mathcal{L}(\Omega|\theta) = \underbrace{\prod_{i\ infected} g\left(D_i - \tau_i \middle| s_i\right)}_{\text{Incubation process (or detection for asymptomatic individuals)}} \underbrace{\underbrace{\prod_{i\ secondary\ case} \lambda_i(\tau_i) e^{-\int_{\tau_1}^{\tau_i} \lambda_i(u)\,\mathrm{d}u}}_{\text{Contribution of infected: Survival until time of infection}} \underbrace{\prod_{i\ not\ infected} e^{-\int_{\tau_1}^{t_{end}} \lambda_i(u)\,\mathrm{d}u}}_{\text{Contribution non infected: Survival until end of household follow-up}}}_{\text{Transmission process}}$$

Incubation process
(or detection for
asymptomatic individuals)

Transmission process

Contribution of infected:
Survival until time of infection

Contribution non infected:
Survival until end of
household follow-up

STATISTICAL INFERENCE

Metropolis-Hastings algorithm:

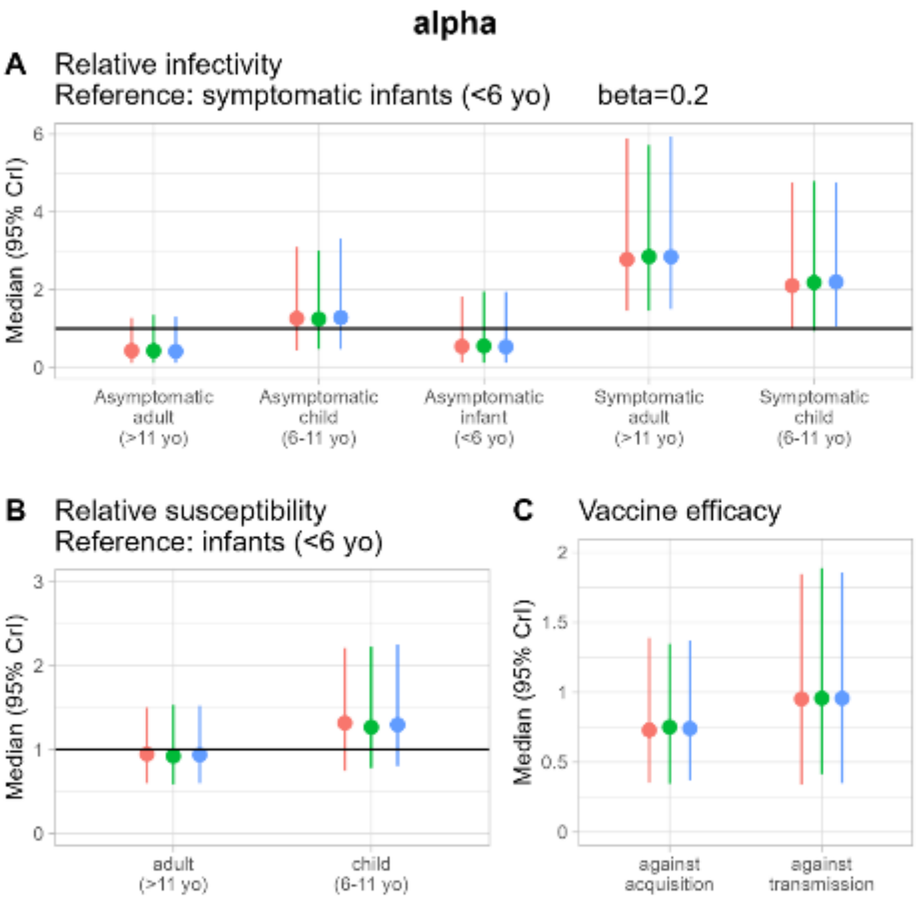
- Update parameters 1 by 1
  - Proposal for delta (natural scale): $new = old + \mathcal{N}(0, \sigma^2)$
  - Proposal for others (log scale): $new = old \, . \, e^{\mathcal{N}(0,\sigma^2)}$
- Update augmented data
  - Proposal symptoms/1st test date: $D_i = d_i + U(0,1)$
  - Proposal infection date: $\tau_i = D_i - rn(g)$

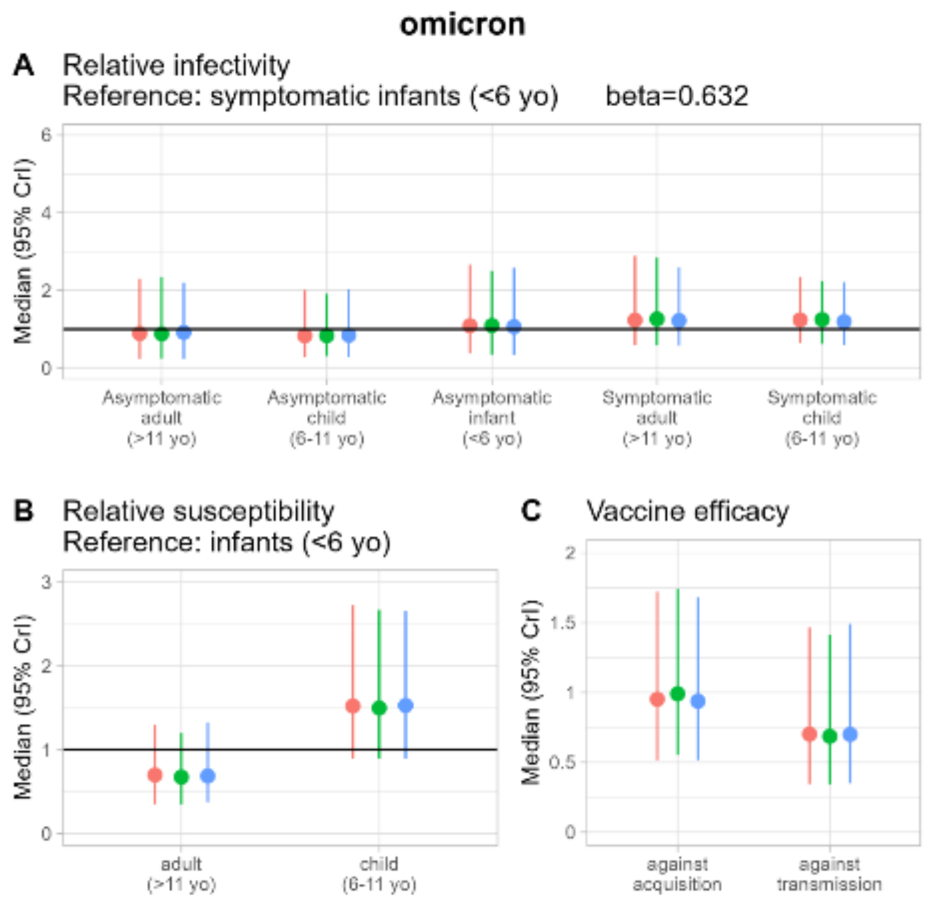
Distributions for priors :

- alpha (risk in community) : unif(0,0.1)
- delta (dependence hh size) : unif(-3,3)
- beta (baseline transmission risk) : unif(0,10)
- μ_i, μ_s (relative infectio and susc) : logN(0,1)
- μ_t, μ_a (protection efficacy) : logN(0,1)

- 500 000 iterations
- Sampling every 500 iterations
- Burn-in of 50 000 iterations
- Evaluation of convergence: visual of chains, probability of acceptance and ess

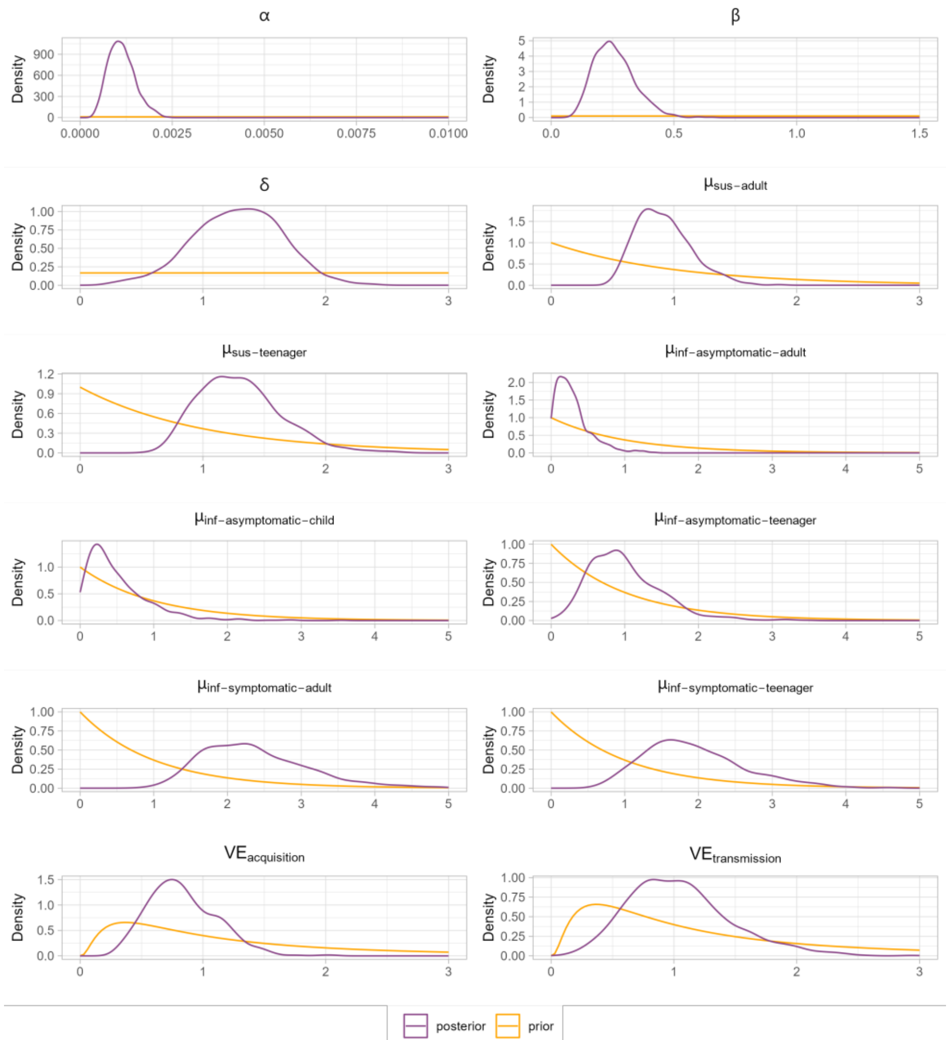
PROTECTION EFFICACY



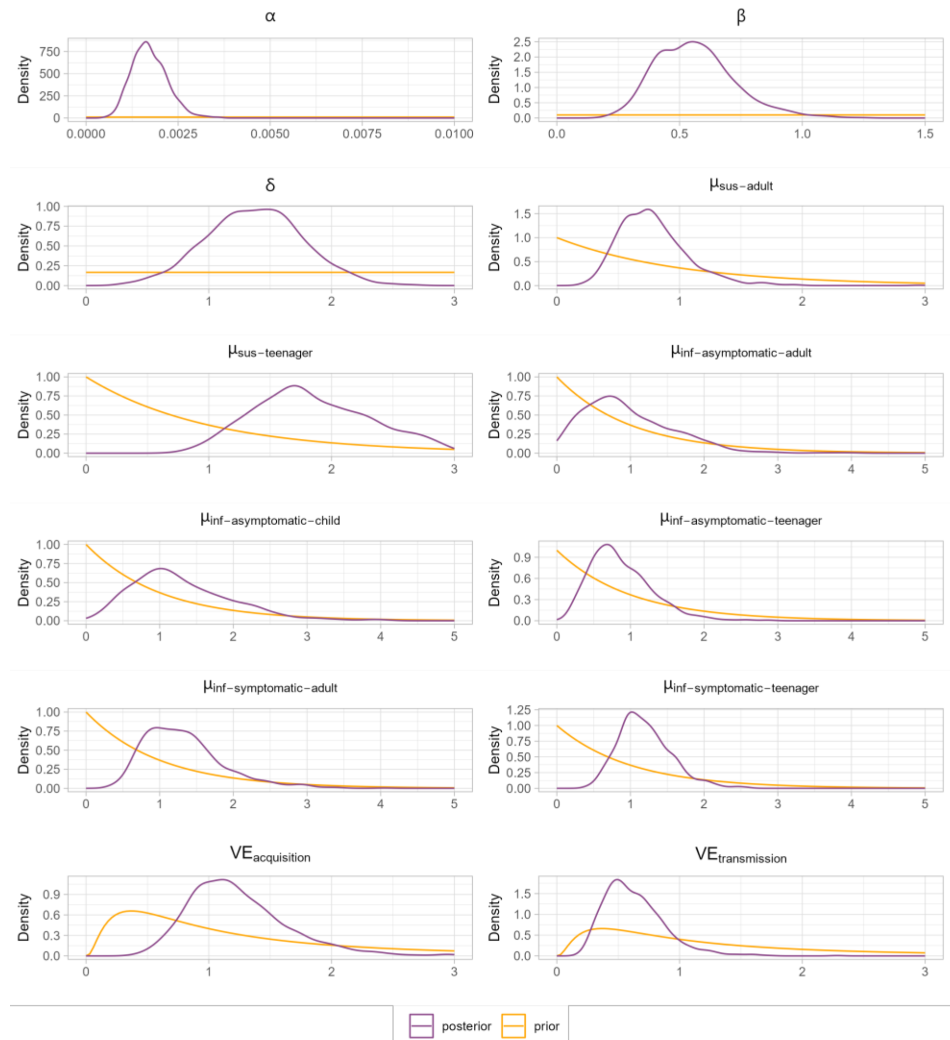


COMPARAISON PRIOR/POSTERIOR

ALPHA



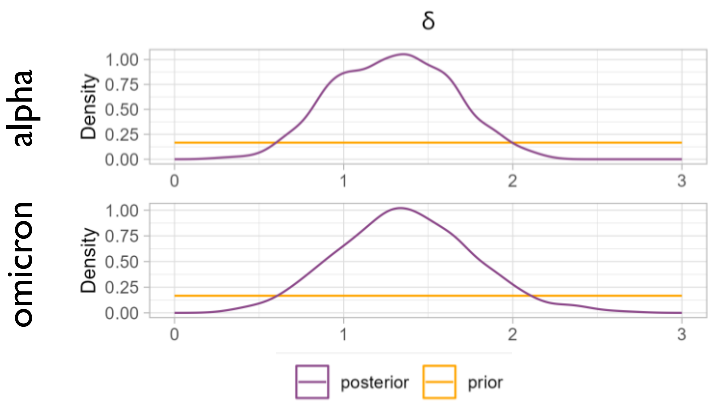
OMICRON



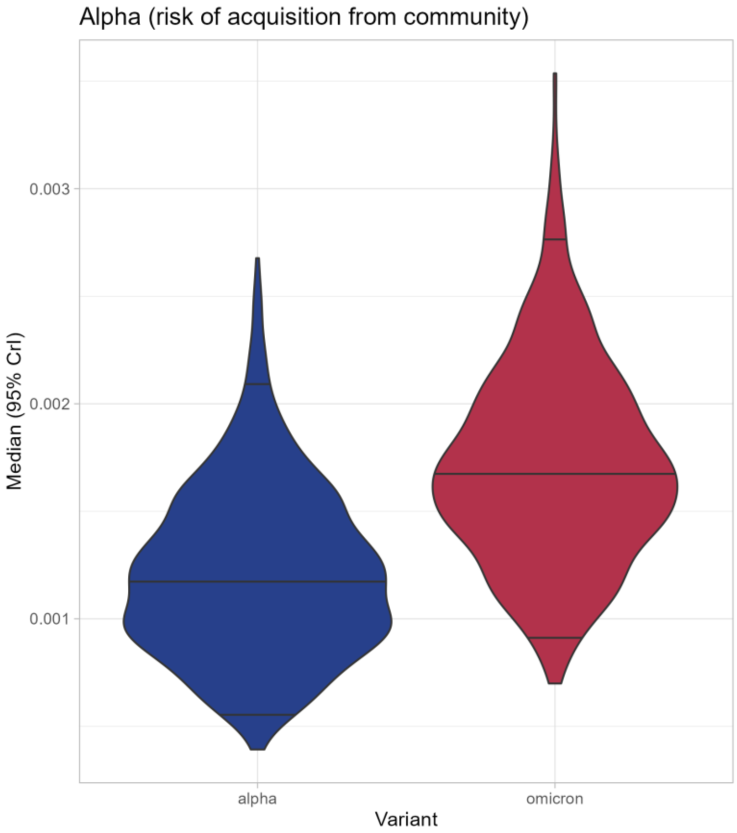
OTHER ESTIMATED PARAMETERS

- Alpha (risk of acquisition from community)

- Delta (dependance of transmission risk on household size)



DELTA estimates (dependance of beta on household size)			
database	median	q2_5	q97_5
alpha	1.30	0.65	1.94
omicron	1.36	0.63	2.20



SIMULATION STUDY

SIMULATION OF OUTBREAKS

- **Choisir un individu index au hasard**
  - Attribuer une date d'infection à 0
  - Déterminer son statut symptomatique (probabilité fixée)
  - Déterminer sa date de symptômes (grâce à la distribution d'incubation)

- **Pour chaque pas de temps t (pendant la durée de suivi) :**
  - Pour tous les individus susceptibles i du foyer, faire :
    - Calculer leur probabilité d'infection $\lambda_i(t)$ avec les paramètres fixés $(\alpha, \beta, \delta, \mu_i^5, \mu_s^2, \mu_a, \mu_t)$
    - Déterminer s'il est infecté ou non

      - Attribuer la date d'infection correspondant à l'itération
      - Déterminer le statut symptomatique
      - Déterminer sa date de symptômes

- **Choisir un cas d'inclusion au hasard :**
  - Lui attribuer (ainsi qu'au reste du foyer) une date d'inclusion en fonction du délai symptômes – inclusion des cas d'inclusion des données

I SIMULATION OF OUTBREAK

p_asympto=0.22 (alpha)
p_asympto=0.25 (omicron)

- **<u>Chose randomly an index individual</u>**
  - Assign an infection date (date_inf)=0
  - Draw u~Unif(0,1)
    If u<pasympto, assign asympto status (infect_status=2)
    Else, infect_status=1 (sympto)

    Sympto or asympto?
  - Assign symptoms onset date with incubation period distributions

- **<u>For each time step t (for duration_followup iterations), do:</u>**
  - <u>For all susceptible individuals i in household, do:</u>
    - Compute proba_inf $\lambda_i(t)$ with fixed parameters $(\alpha, \beta, \delta, \mu_i^5, \mu_S^2, \mu_a, \mu_t)$

      $$\lambda_i(t) = \alpha + \sum_{\substack{j \in H_i, \\ \tau_j < t}} h_{j \to i}(t)$$
    - Draw u~Unif(0,1).
      If u<proba_inf,

      Infecté ou non ?

      - Assign date of infection corresponding to the iteration.
      - Draw u~Unif(0,1).
        If u<pasympto, assign infect_status=2 (asympto).
        Else, infect_status=1 (sympto).

        Sympto ou asympto ?
      - Assign symptoms onset date date_sympt with incubation period distributions
    - Else, infect_status=0.

- **<u>Chose randomly an inclusion case:</u>**
  - Assign inclusion date from delay symptoms–inclusion of inclusion cases from Pedcovid

Alpha (84 foyers) :	**Omicron (46 foyers) :**
67 incluCase sympto	36 incluCase sympto
53 incluCase aussi index	37 incluCase aussi index

Scénarios d'inclusion des foyers

Recrutement via **enfants** (PedCovid-like)

- Si le cas d'inclusion est un adulte → exclusion
- Si le cas d'inclusion est un enfant :
  - Si symptômes après inclusion → exclusion
  - Sinon → recrutement
    (reflétant les nombres de symptomatiques et cas index des cas d'inclusion de PedCovid)

Recrutement via **adultes** (Via adults)

- Si le cas d'inclusion est un **enfant** → exclusion
- Si le cas d'inclusion est un **adulte** :
  - Si symptômes après inclusion → exclusion
  - Sinon → recrutement
    (reflétant les nombres de symptomatiques et cas index des cas d'inclusion de PedCovid)

Recrutement **aléatoire** (Random)

- Pas de classification/exclusion
- Tirer un nombre total au hasard

Sur-sampling avec recrutement via enfants (PedCovid-like + sampling)

- Si le cas d'inclusion est un adulte → exclusion
- **Si aucun adulte infecté dans le foyer → exclusion**
- Si le cas d'inclusion est un enfant :
  - Si symptômes après inclusion → exclusion
  - Sinon → recrutement
    (reflétant les nombres de symptomatiques et cas index des cas d'inclusion de PedCovid)

Résultats hypothèse nulle

Période alpha

alpha = 0,001
beta = 0,2
delta = 1,3
rel_inf = 1
rel_susc = 1
VE_acq = 0,8
VE_transm = 1

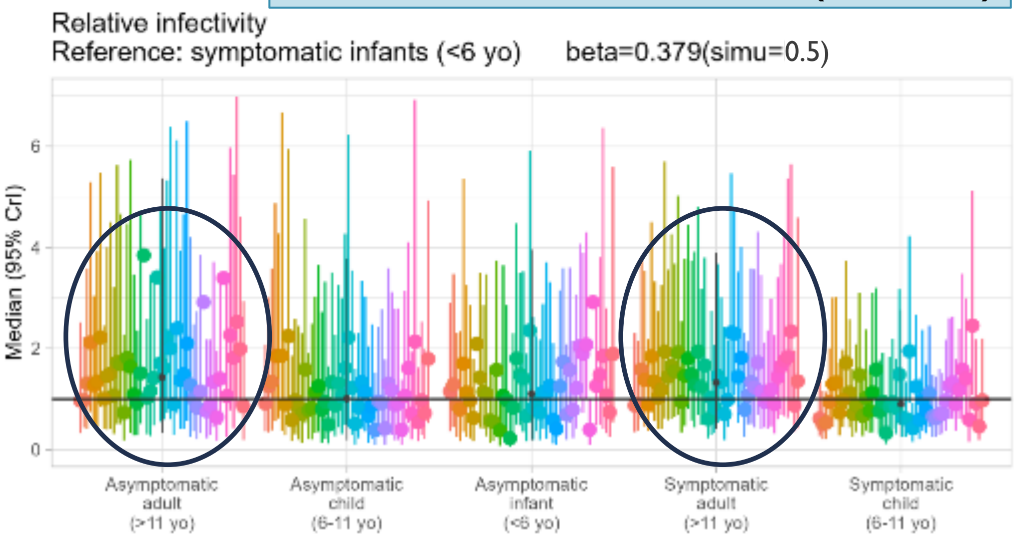
Période omicron

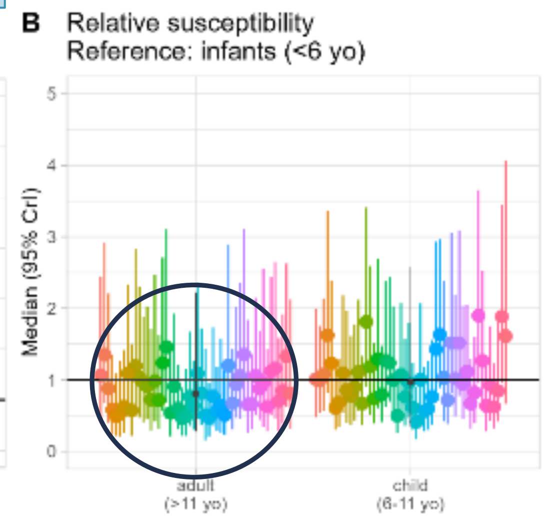
alpha = 0,002
beta = 0,5
delta = 1,4
rel_inf = 1
rel_susc = 1
VE_acq = 0,8
VE_transm = 0,8

Résultats hypothèse nulle

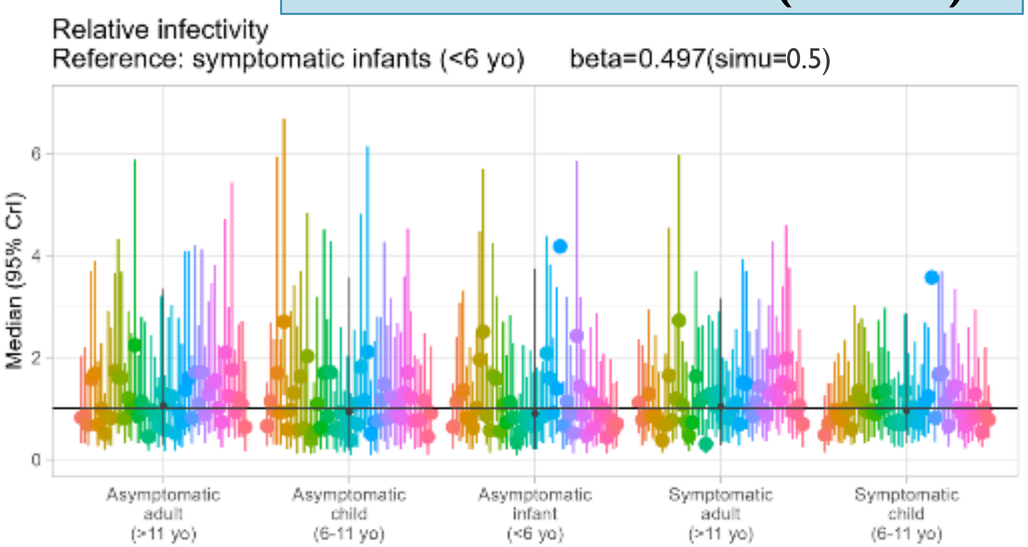
Infectiosité relative = 1
Susceptibilité relative = 1

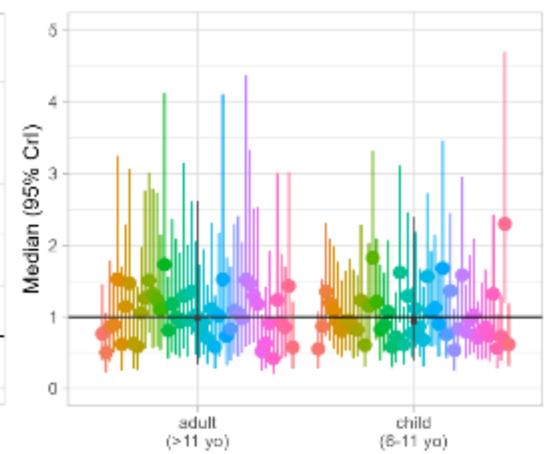
Recrutement PedCovid-like (omicron)





Recrutement aléatoire (omicron)



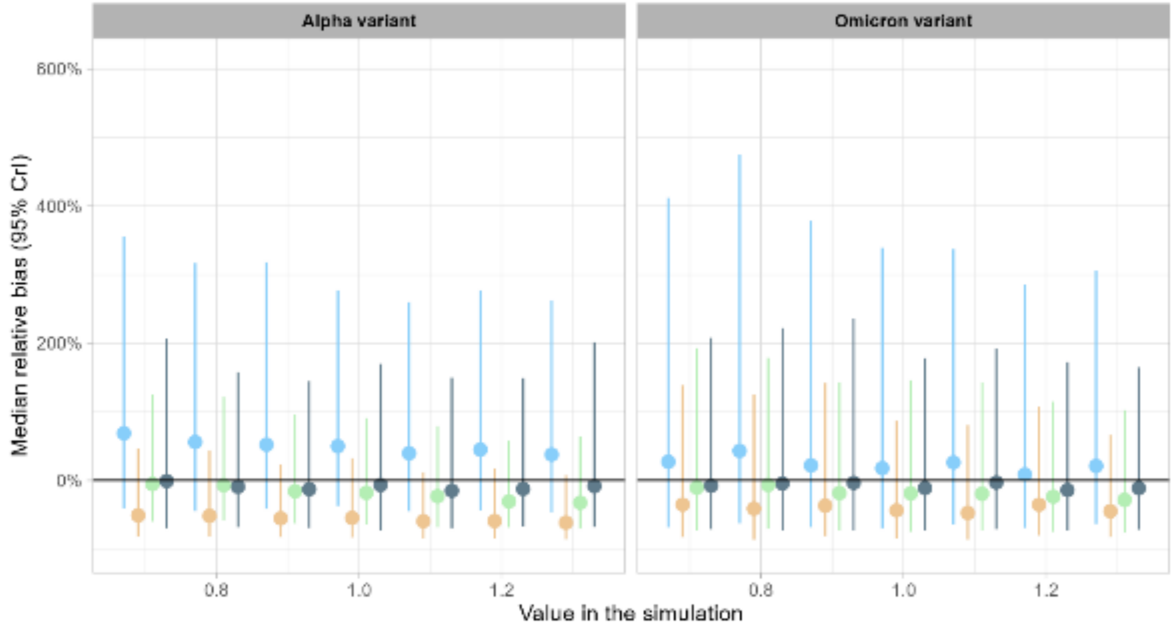


Recrutement PedCovid-like :
Sur-estimation de l'infectiosité des adultes
Sous-estimation de la susceptibilité des adultes

Recrutement aléatoire :
Bonne estimation

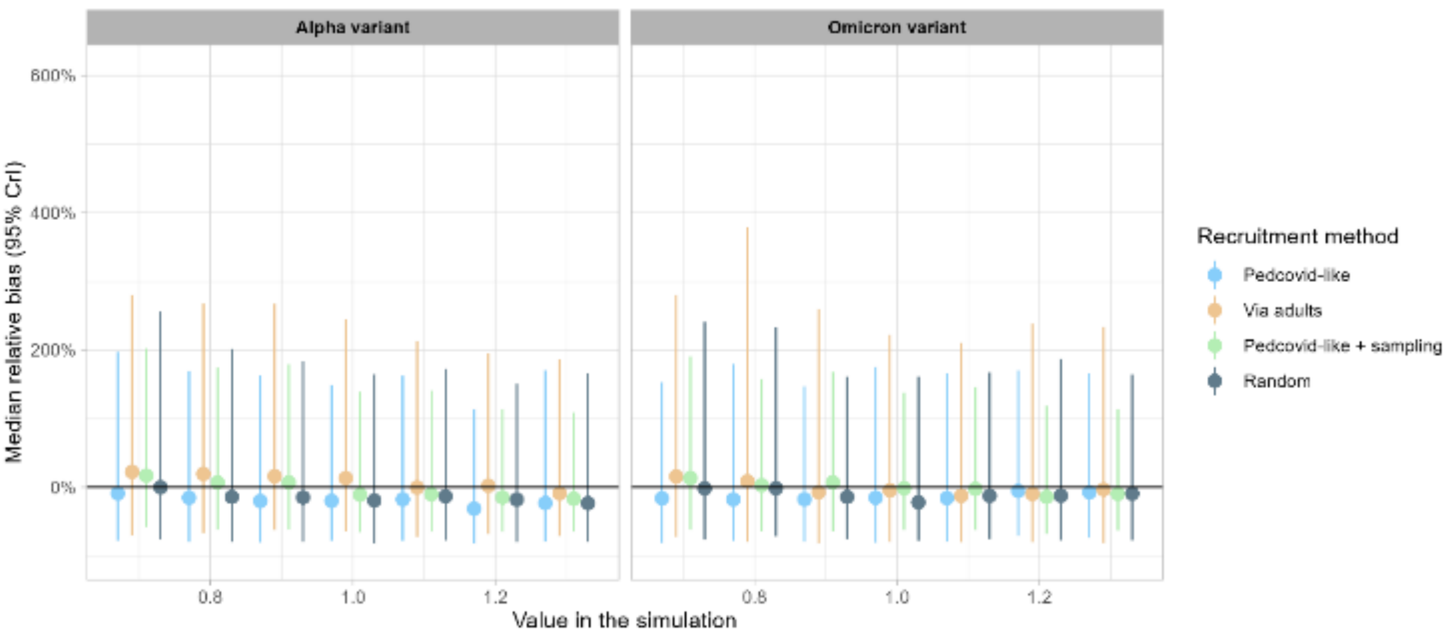
Bias for Infectivity of Symptomatic Adults

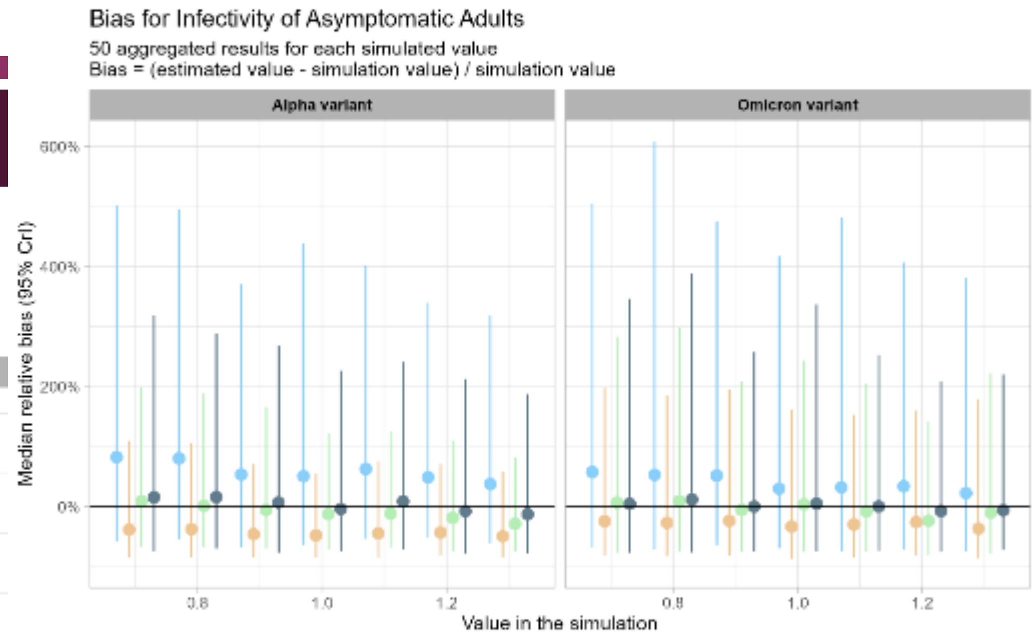
50 aggregated results for each simulated value
Bias = (estimated value - simulation value) / simulation value

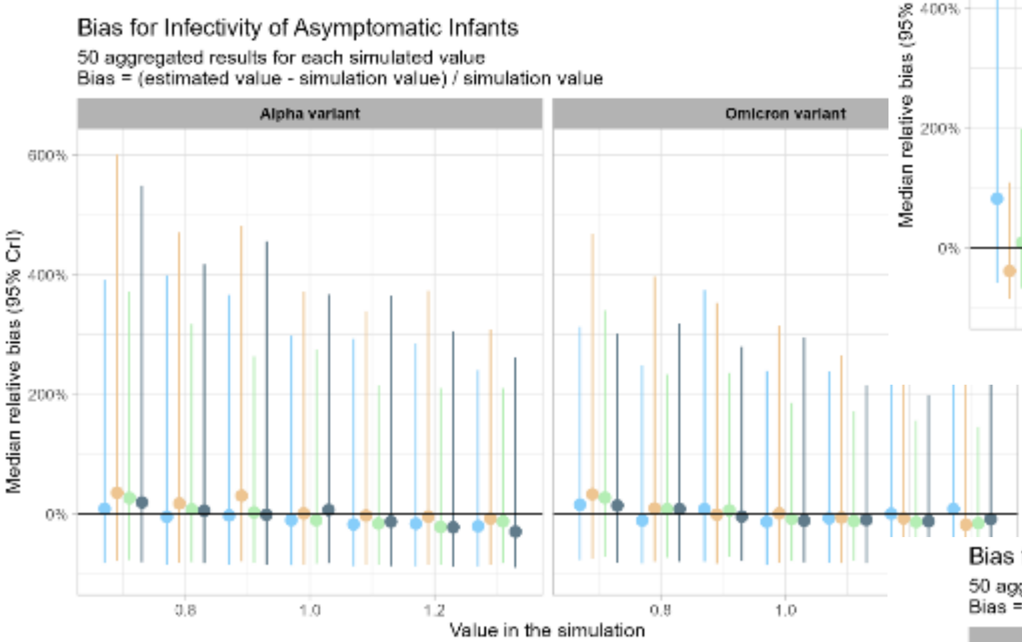


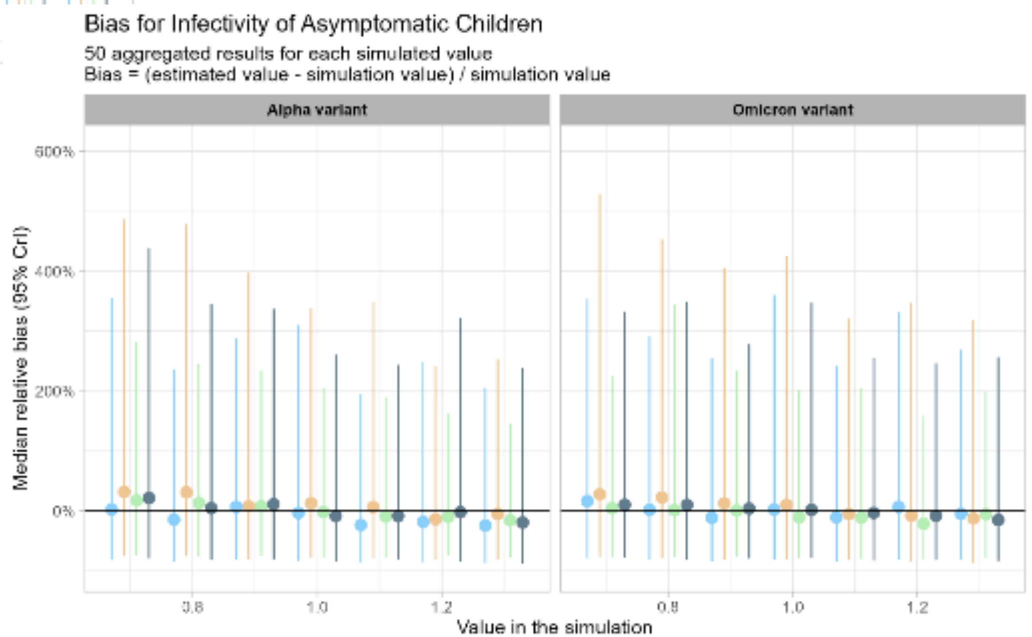
Bias for Infectivity of Symptomatic Children

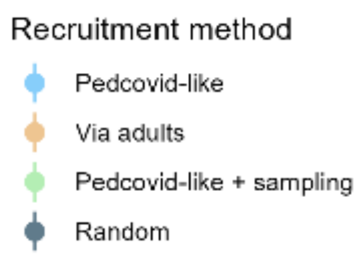
50 aggregated results for each simulated value
Bias = (estimated value - simulation value) / simulation value




Bias for Infectivity of Asymptomatic Adults
50 aggregated results for each simulated value
Bias = (estimated value - simulation value) / simulation value
Alpha variant
Omicron variant
Median relative bias (95% CrI)
600%
400%
200%
0%
0.8
1.0
1.2
Value in the simulation

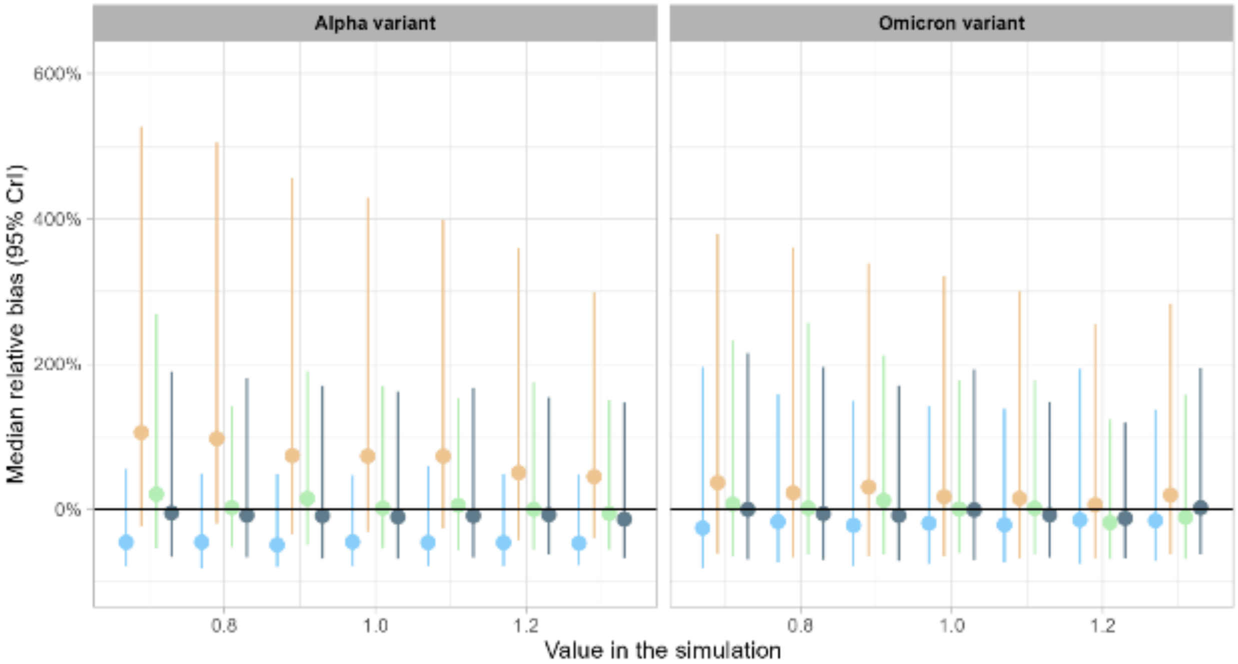

Bias for Infectivity of Asymptomatic Infants
50 aggregated results for each simulated value
Bias = (estimated value - simulation value) / simulation value
Alpha variant
Omicron variant
Median relative bias (95% CrI)
600%
400%
200%
0%
0.8
1.0
1.2
Value in the simulation


Bias for Infectivity of Asymptomatic Children
50 aggregated results for each simulated value
Bias = (estimated value - simulation value) / simulation value
Alpha variant
Omicron variant
Median relative bias (95% CrI)
600%
400%
200%
0%
0.8
1.0
1.2
Value in the simulation


Recruitment method
Pedcovid-like
Via adults
Pedcovid-like + sampling
Random

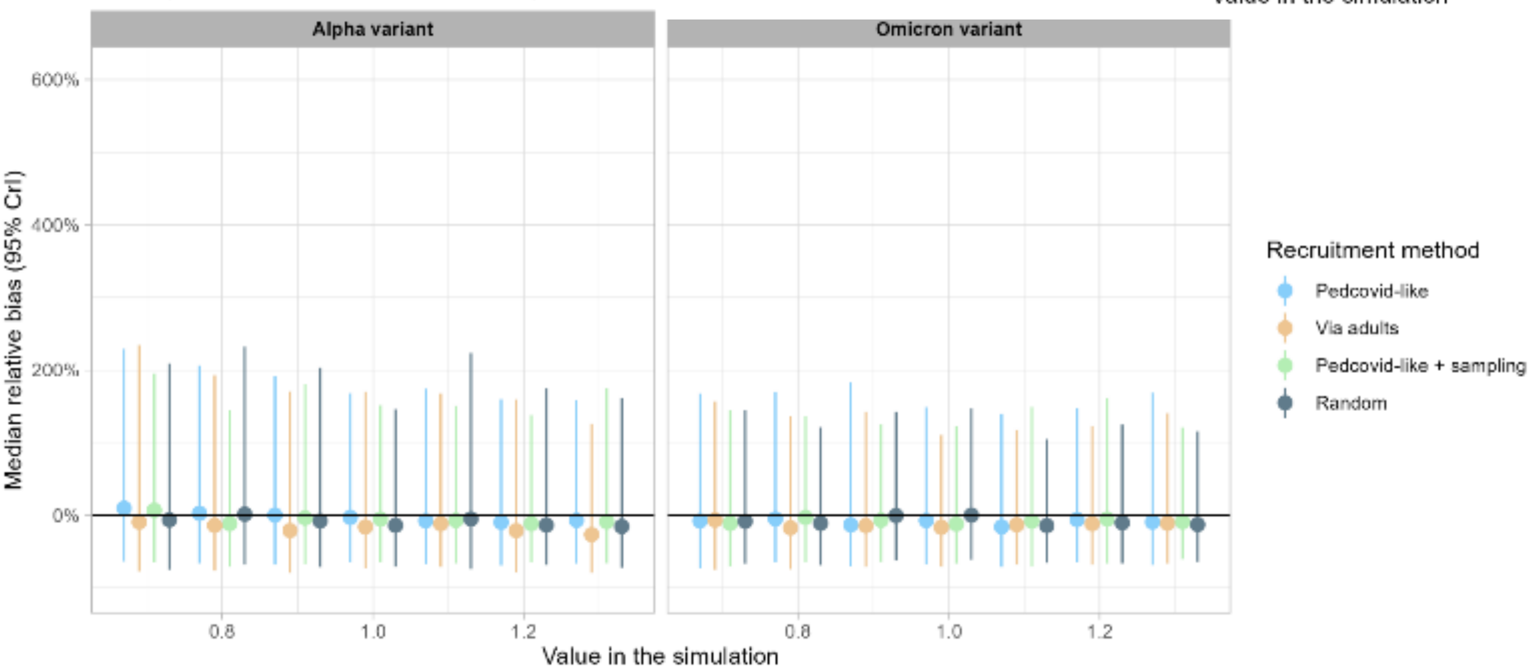
Bias for Susceptibility of Adults

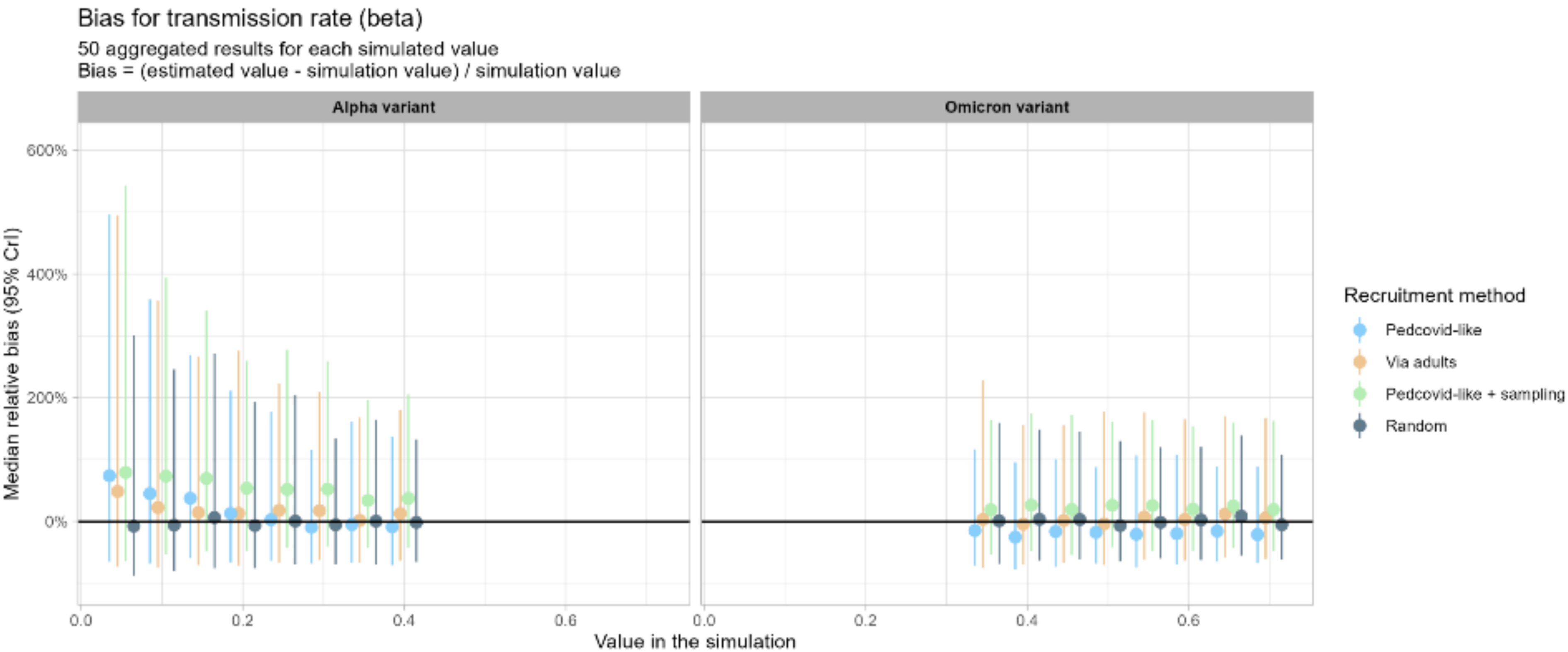
50 aggregated results for each simulated value
Bias = (estimated value - simulation value) / simulation value



Bias for Susceptibility of Children

50 aggregated results for each simulated value
Bias = (estimated value - simulation value) / simulation




Bias for transmission rate (beta)
50 aggregated results for each simulated value
Bias = (estimated value - simulation value) / simulation value
Alpha variant
Omicron variant
600%
400%
200%
0%
Median relative bias (95% CrI)
0.0
0.2
0.4
0.6
Value in the simulation
Recruitment method
Pedcovid-like
Via adults
Pedcovid-like + sampling
Random

SENSIBILITY ANALYSIS ON MU_INF AND MU_SUSC PRIORS

SENSIB ON MU_INF AND MU_SUSC

ALPHA PERIOD

	Exp(1)					logN(0,1)				
	Mean	Median	CrI ETI 95%	Best	Mode	Mean	Median	CrI ETI 95%	Best	Mode
alpha - prior										
prior	1	0.69	0.03-3.68	-	0	1.65	1	0.14-7.10	-	0.37
alpha - susceptibility										
rSC	1.3	1.27	0.78-2.07	0.56	1.15	1.27	1.23	0.76-2.01	1.74	1.15
rSA	0.92	0.89	0.57-1.41	0.65	0.79	0.9	0.87	0.56-1.35	1.03	0.84
alpha - infectiousness										
rInfAA	0.29	0.24	0.01-0.92	0.55	0.13	0.42	0.35	0.1-1.13	0.17	0.24
rInfAC	1	0.92	0.28-2.21	0.54	0.89	1.25	1.13	0.36-3	0.88	0.9
rInfAI	0.51	0.38	0.03-1.62	0.88	0.22	0.65	0.55	0.12-1.78	0.5	0.41
rInfSA	2.38	2.28	1.29-4.12	2.37	2.22	2.93	2.71	1.41-5.57	2.66	2.3
rInfSC	1.94	1.85	0.9-3.56	2.04	1.62	2.41	2.19	1-5.26	1.89	1.99
alpha - logLikelihood										
logLik	-1404.96	-1404.57	-1421.34--1390.12	-1434.3	-1403.64	-1404.52	-1404.54	-1421.8--1388.9	-1430.11	-1405.91

ETI = equal-tailed interval (quantiles 2,5% and 97,5%) → 95% of points fall in this interval
Best = param that gives highest likelihood
Mode = max of the density curve

SENSIB ON MU_INF AND MU_SUSC

OMICRON PERIOD

	Exp(1)					logN(0,1)				
	Mean	Median	CrI ETI 95%	Best	Mode	Mean	Median	CrI ETI 95%	Best	Mode
omicron - prior										
prior	1	0.69	0.03-3.68	-	0	1.65	1	0.14-7.10	-	0.37
omicron - susceptibility										
rSC	1.89	1.8	1.04-3.17	0.9	1.69	1.96	1.86	1.09-3.39	1.11	1.73
rSA	0.78	0.74	0.36-1.45	0.8	0.74	0.8	0.77	0.39-1.46	0.43	0.66
omicron - infectiousness										
rInfAA	0.96	0.84	0.11-2.26	1.81	0.72	1.07	0.94	0.2-2.77	1.13	0.76
rInfAC	0.87	0.79	0.26-1.88	1.6	0.68	0.91	0.83	0.29-1.98	0.42	0.68
rInfAI	1.31	1.19	0.3-2.97	2.24	1.01	1.45	1.34	0.38-3.19	1.57	1.26
rInfSA	1.33	1.25	0.57-2.7	0.75	0.96	1.38	1.28	0.56-2.67	1.25	1.16
rInfSC	1.2	1.15	0.61-2.05	0.65	1.02	1.23	1.18	0.61-2.14	1.37	1.13
omicron - logLikelihood										
logLik	-730.16	-729.84	-744.02--717.53	-753.54	-729.39	-729.86	-729.68	-743.85--718.14	-752.6	-730.05

ETI = equal-tailed interval (quantiles 2,5% and 97,5%) → 95% of points fall in this interval
Best = param that gives highest likelihood
Mode = max of the density curve

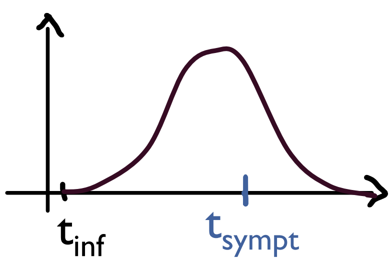
SENSIBILITY ANALYSIS ON GENERATION TIMES

SENSIB ON GENERATION TIMES

Baseline model:

- Prior as is:
  alpha: unif(0, 0.1)
  delta: unif(-3,3)
  beta: unif(0,10)
  mu_protect: logN(0,1)
  mu_inf: logN(0,1)
  mu_susc: logN(0,1)
- **Generation time:** an der Heiden + Chen without truncation

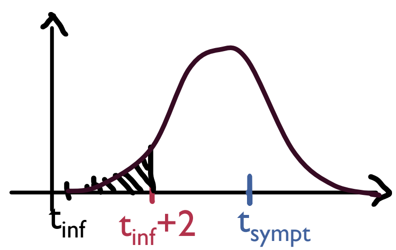
→ No truncation



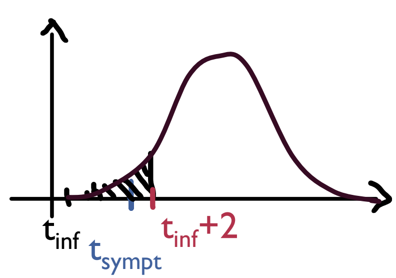
Option 1

→ No infectivity for 1st 2 days for alpha and 1st day for omicron (not dependent on incubation time)

Cas incub > 2



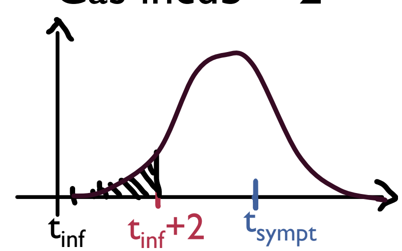
Cas incub < 2



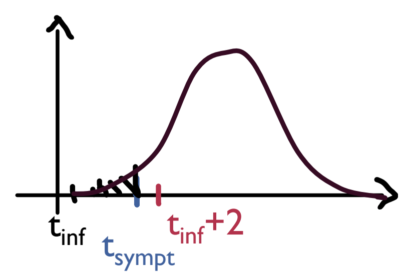
Option 2

→ No infectivity for 1st 2 days for alpha and 1st day for omicron or until beginning of symptoms if incubation is small

Cas incub > 2

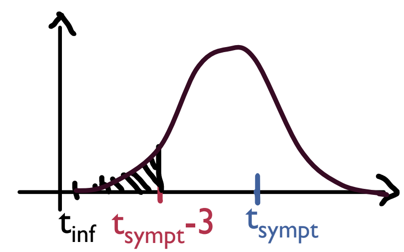


Cas incub < 2



Option 3

→ No infectivity before 3 days before symptoms for alpha and before 2 days before symptoms for omicron (or from infection if incub is small)

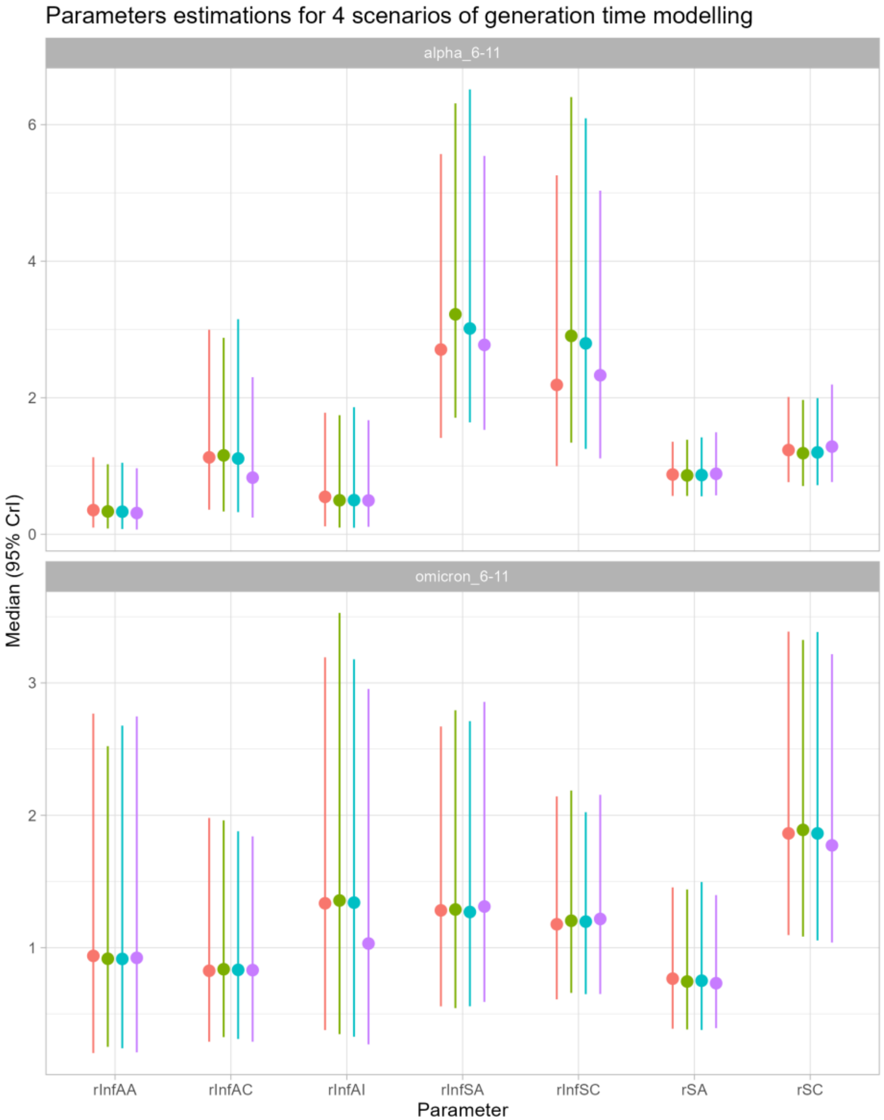


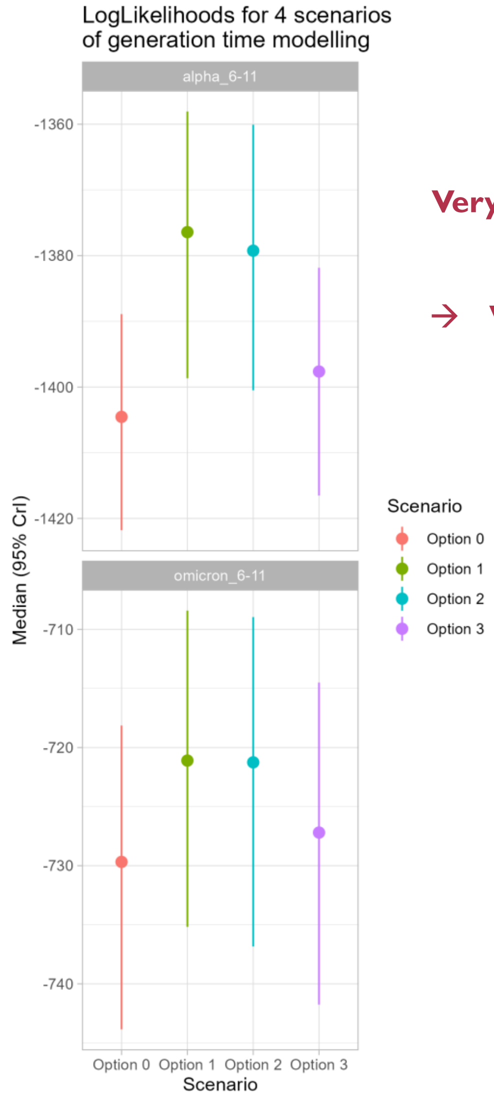
SENSIB ON GENERATION TIMES

Characteristic	**Option 0**, N = 901[1]	**Option 1**, N = 901[1]	**Option 2**, N = 901[1]	**Option 3**, N = 901[1]
ALPHA period				
logLik	-1,405 (-1,422, -1,389)	-1,376 (-1,399, -1,358)	-1,379 (-1,400, -1,360)	-1,398 (-1,417, -1,382)
rSC	1.23 (0.76, 2.01)	1.19 (0.71, 1.97)	1.20 (0.72, 1.99)	1.28 (0.76, 2.19)
rSA	0.87 (0.56, 1.35)	0.86 (0.56, 1.38)	0.87 (0.56, 1.42)	0.89 (0.57, 1.49)
rInfAA	0.35 (0.10, 1.13)	0.33 (0.08, 1.03)	0.33 (0.08, 1.05)	0.31 (0.07, 0.97)
rInfAC	1.13 (0.36, 3.00)	1.16 (0.33, 2.88)	1.11 (0.32, 3.15)	0.83 (0.24, 2.30)
rInfAI	0.55 (0.12, 1.78)	0.50 (0.10, 1.74)	0.50 (0.10, 1.86)	0.49 (0.11, 1.67)
rInfSA	2.71 (1.41, 5.57)	3.22 (1.71, 6.31)	3.01 (1.64, 6.52)	2.77 (1.53, 5.54)
rInfSC	2.19 (1.00, 5.26)	2.91 (1.34, 6.40)	2.80 (1.25, 6.09)	2.33 (1.11, 5.03)
OMICRON period				
logLik	-730 (-744, -718)	-721 (-735, -708)	-721 (-737, -709)	-727 (-742, -715)
rSC	1.86 (1.09, 3.39)	1.89 (1.08, 3.32)	1.86 (1.05, 3.38)	1.77 (1.04, 3.22)
rSA	0.77 (0.39, 1.46)	0.74 (0.38, 1.44)	0.75 (0.38, 1.50)	0.73 (0.39, 1.40)
rInfAA	0.94 (0.20, 2.77)	0.92 (0.25, 2.52)	0.92 (0.24, 2.68)	0.92 (0.21, 2.75)
rInfAC	0.83 (0.29, 1.98)	0.84 (0.32, 1.96)	0.83 (0.31, 1.88)	0.83 (0.29, 1.84)
rInfAI	1.34 (0.38, 3.19)	1.36 (0.35, 3.53)	1.34 (0.33, 3.18)	1.03 (0.27, 2.95)
rInfSA	1.28 (0.56, 2.67)	1.29 (0.54, 2.79)	1.27 (0.56, 2.71)	1.31 (0.59, 2.86)
rInfSC	1.18 (0.61, 2.14)	1.20 (0.66, 2.19)	1.20 (0.65, 2.02)	1.22 (0.65, 2.15)

[1] Median (eti_low, eti_high)

SENSIB ON GENERATION TIMES





Very similar values for each option, loglikelihood also similar.

→ **We choose the one with lowest loglikelihood, and the simplest**

→ **Option 0 (baseline, no truncation)**